Highlights

Optimizing Irrigation and Fertilization Strategies for Crop Growth: A Comparative Study of Genetic Algorithm and Model Predictive Control under Weather Uncertainty

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- Coupled ODE crop model with FIR convolution for delayed absorption and EMA filtering for cumulative stress
- Genetic algorithm optimization achieves 35% revenue improvement over farmer best practices, on average
- Model predictive control with Bayesian-optimized parameters for adaptive daily resource allocation
- Four-way strategy comparison across 21 stochastic weather scenarios from normal to extreme
- Key finding: pre-optimization outperforms adaptive control; MPC offers sustainability advantages

Optimizing Irrigation and Fertilization Strategies for Crop Growth: A Comparative Study of Genetic Algorithm and Model Predictive Control under Weather Uncertainty

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Abstract

Rising production costs and increasing weather variability pose significant challenges to agricultural profitability, motivating computational approaches to optimize resource allocation. This paper presents a coupled ordinary differential equation (ODE) crop model that captures two key physiological phenomena often neglected in optimization studies: delayed nutrient absorption and cumulative stress tracking, which we address via finite impulse response (FIR) convolution and exponential moving average (EMA) filtering. We compare two optimization approaches for irrigation and fertilizer scheduling: genetic algorithm (GA) optimization for fixed seasonal strategies and model predictive control (MPC) for adaptive daily decision-making. Applied to corn production in Iowa across 21 stochastic weather scenarios spanning normal conditions to extreme drought and heat stress, our results yield a surprising conclusion. GA optimization achieves 35% higher revenue than farmer best practices ($842 vs. $626 per acre), and contrary to our initial hypothesis, the fixed GA strategies outperform adaptive MPC in all 21 scenarios. The GA trained under normal weather conditions achieves the highest mean revenue with the lowest risk (coefficient of variation 14.7%), while MPC, despite its daily adaptation, cannot match the GA's full-season optimization horizon. However, MPC offers a sustainability advantage: it achieves competitive returns while using dramatically less water and fertilizer than all other strategies. The practical conclusion is that for seasonal

agricultural planning, pre-optimization for expected conditions outperforms adaptive control, and the choice of which GA strategy to deploy depends simply on whether extreme weather is anticipated.

Keywords: precision agriculture, resource optimization, crop growth model, genetic algorithm, cumulative stress tracking, model predictive control

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1. Introduction

The agriculture sector in the United States faces significant challenges as the number of farms declines and the cost of farming continues to rise [1]. Rising production expenses for equipment, seeds, and labor, coupled with elevated interest rates and declining commodity prices, have made farming increasingly expensive. To navigate this challenging landscape, farmers are employing strategies such as cost management and operational optimization. One promising approach is to use modeling and simulation to optimize farm operations without substantial capital investment. Recent advances in computational methods have enabled sophisticated digital-twin frameworks for precision agriculture [2, 3], and machine learning techniques have been applied to optimize sensor placement and resource delivery in agricultural systems [4, 5].

Mathematical modeling of crop growth has a rich history, with models ranging from simple empirical relationships to complex mechanistic simulations. Logistic growth models, first proposed by Verhulst in 1838, remain widely used due to their interpretability and ability to capture resource-limited growth dynamics [6]. More sophisticated crop models such as DSSAT [7], APSIM [8], and WOFOST [9] simulate detailed physiological processes but require extensive parameterization and may be computationally expensive for optimization applications. In contrast, reduced-order models that capture essential dynamics while remaining tractable for optimization have gained attention in precision agriculture [10].

Optimization of irrigation and fertilizer application has been studied using various approaches, including linear and nonlinear programming [11], dynamic programming [12], and metaheuristic algorithms [13]. Genetic algorithms (GAs) are particularly well-suited for this domain because they can handle nonlinear, non-convex objective functions and do not require gradient information [14]. Previous work has applied GAs to irrigation scheduling [15] and fertilizer optimization [16], but these studies often use simplified

plant response models that do not capture the delayed, cumulative effects of resource application.

Climate change is increasing the frequency and severity of extreme weather events, posing significant challenges to agricultural production [17]. Traditional approaches to irrigation and fertilizer scheduling rely on fixed strategies—either following agronomic best practices or strategies optimized for expected conditions—that cannot adapt when actual weather deviates from assumptions. As drought, heat waves, and other extremes become more common, there is growing interest in adaptive resource management strategies that can respond to observed conditions in real time.

Model Predictive Control (MPC) offers an alternative approach: a closed-loop control strategy that repeatedly solves an optimization problem over a finite horizon, applies only the first control action, then re-optimizes based on updated state and disturbance information [18]. This receding-horizon structure enables the controller to adapt to changing conditions while maintaining optimality over the planning horizon. The central question motivating this work is whether MPC's adaptive capabilities can outperform fixed strategies that are pre-optimized for specific conditions.

This paper makes the following contributions:

1. A coupled ODE crop growth model that captures delayed nutrient absorption via FIR convolution and cumulative stress tracking via EMA filtering, providing a physiologically-motivated yet computationally tractable framework for optimization.
2. A genetic algorithm approach that optimizes fixed irrigation and fertilizer strategies for specific weather conditions, achieving 35% revenue improvement over farmer best practices, on average.
3. An MPC framework with Bayesian-optimized parameters that adapts resource allocation daily based on observed plant state and weather forecasts.
4. A stochastic weather scenario generator that creates realistic test conditions ranging from normal variability to extreme drought and heat stress.
5. A comprehensive comparison of four strategies—farmer best practices, GA optimized for normal conditions, GA optimized for drought, and adaptive MPC—across 21 weather scenarios, revealing that pre-optimization outperforms adaptive control for seasonal agricultural planning.

The remainder of this paper is organized as follows. Section 2 presents the

coupled ODE crop model with FIR convolution and EMA filtering. Section 3 describes the stochastic weather scenario generation process. Sections 4 and 5 detail the genetic algorithm and model predictive control approaches, respectively. Section 6 presents results from the four-way strategy comparison across 21 scenarios. Section 7 interprets the findings and discusses limitations. Section 8 concludes with practical recommendations.

2. Generalized, coupled-ODE crop model

This paper adopts a control systems perspective on crop growth, treating the plant as a dynamical system with internal states $\mathbf{x} = [x_h, x_A, x_N, x_c, x_P]^\top$, comprising plant height (m), leaf area (m^2), number of leaves (count), flower spikelet count, and fruit biomass (kg). Control inputs $\mathbf{u} = [u_w, u_f]^\top$, irrigation (in) and fertilizer (lbs), are decision variables chosen by the farmer. Disturbance inputs $\mathbf{d} = [d_S, d_T, d_R]^\top$, precipitation (in), temperature (°C), and radiation (W/m^2), affect the system but cannot be directly controlled. The internal dynamics follow logistic growth with time-varying growth rates and carrying capacities that depend on environmental conditions and resource availability. The output is a function of the state of the crop at the end of the season. This framing, visualized in Figure 1, naturally motivates an optimization question: how can we choose an irrigation and fertilizer strategy that optimizes crop revenue at the end of the growing season? We consider two optimal control paradigms: open-loop control via genetic algorithm, which pre-computes a fixed input trajectory, and closed-loop model predictive conrol, which adapts inputs based on state feedback.

2.1. Effects of inputs on plant growth

Different inputs affect different aspects of plant growth, and such relationships are crop dependent. In this study, we use corn to demonstrate the framework. For corn, Tables 1 and 2 summarize these relationships based on agronomic literature [19, 20, 21]. We note that fertilizer-assisted growth encourages a taller stalk and larger leaf area via both increased carrying capacity and higher growth rate. Irrigation-assisted growth encourages larger leaf area, number of tassels (flower size), and ear size (fruit biomass) via increased carrying capacity.

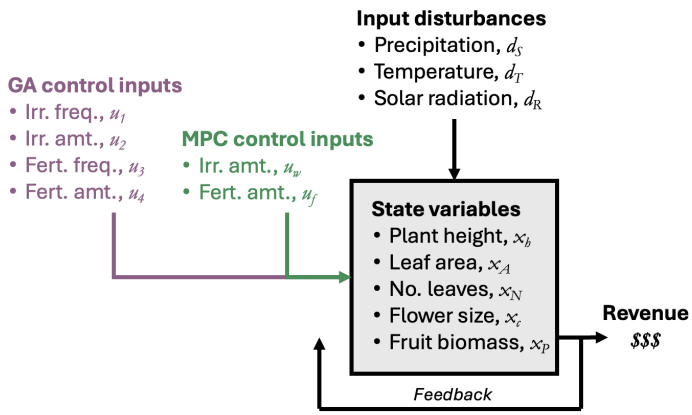

Figure 1: This paper treats a crop as a control system with its own internal state variables and dynamics, affected by input controls, input disturbances, and feedback. Output is a function of the state variables.

2.2. Growth dynamics

In our approach, each state variable follows logistic growth with time-varying growth rates and carrying capacities. These parameters are modulated by nutrient factors—dimensionless quantities in $[0, 1]$ that translate cumulative input history into growth effects, as detailed in Section 2.5. The logistic growth takes the general form:

$$\frac{dx}{dt} = \hat{a}_x(t) \cdot x(t) \left(1 - \frac{x(t)}{\hat{k}_x(t)}\right) \tag{1}$$

where $\hat{a}_x(t)$ is the effective growth rate and $\hat{k}_x(t)$ is the effective carrying capacity, both functions of the nutrient factors.

The effective parameters are proportional to their typical values by the geometric mean of the relevant nutrient factors, reflecting multiplicative rather than additive effects. This choice is motivated by the observation that growth rates compound over time, making geometric averaging appropriate [27].

Plant height growth rate and carrying capacity are modulated by fertilizer, temperature, and radiation through their respective nutrient factors as follows:

$$\hat{a}_h(t) = a_h(\nu_f \nu_T \nu_R)^{1/3}, \quad \hat{k}_h(t) = k_h(\nu_f \nu_T \nu_R)^{1/3} \tag{2}$$

State variable	Irrigation on growth rate	Fertilizer on growth rate	Irrigation on capacity	Fertilizer on capacity
Plant height h	$\sim$	$+$	$\sim$	$+$
Leaf area A	$\sim$	$+$	$+$	$+$
Number of leaves N	$\sim$	$\sim$	$\sim$	$\sim$
Flower size c	$\sim$	$\sim$	$+$	$\sim$
Fruit biomass P	$\sim$	$\sim$	$+$	$+$

Table 1: Effects of irrigation and fertilizer on growth dynamics. "+" indicates positive effect, "$\sim$" indicates a negligible effect. [22], [23], [24], [25].

State variable	Temp. on growth rate	Temp. on capacity	Radiation on growth rate	Radiation on capacity
Plant height h	$+$	$+$	$+$	$+$
Leaf area A	$+$	$+$	$+$	$+$
Number of leaves N	$\sim$	$+$	$\sim$	$+$
Flower size c	$-$	$-$	$-$	$-$
Fruit biomass P	$+$	$+$	$+$	$+$

Table 2: Effects of temperature and solar radiation on growth dynamics. "+" indicates positive effect, "$\sim$" indicates a negligible effect, "–" indicates a negative effect. For flower size, excess heat and radiation reduce flower development, hence negative effects. [26]

where a_h is the typical plant height growth rate and k_h is the typical plant height carrying capacity, both chosen to reflect what is observed in corn development literature.

Leaf area responds to fertilizer, temperature, radiation, and, additionally, to water. Leaf area is also coupled to plant height, representing vegetative growth. We capture those relationships with

$$\hat{a}_A(t) = a_A(\nu_f \nu_T \nu_R)^{1/3}, \quad \hat{k}_A(t) = k_A \left(\nu_w \nu_f \nu_T \nu_R \frac{\hat{k}_h}{k_h} \right)^{1/5} \tag{3}$$

where a_A and k_A, respectively, are the growth rate and carrying capacity both chosen to reflect typical literature values.

Number of leaves depends only on temperature and radiation through the carrying capacity:

$$\hat{a}_N(t) = a_N, \quad \hat{k}_N(t) = k_N(\nu_T \nu_R)^{1/2} \tag{4}$$

Flower size (spikelet count) exhibits inverse dependence on temperature and radiation, as excess heat and light reduce flowering:

$$\hat{a}_c(t) = a_c \left(\frac{1}{\nu_T} \frac{1}{\nu_R} \right)^{1/2}, \quad \hat{k}_c(t) = k_c \left(\nu_w \frac{1}{\nu_T} \frac{1}{\nu_R} \right)^{1/3} \tag{5}$$

Fruit biomass depends on all inputs and is coupled to both vegetative growth and flowering:

$$\hat{a}_P(t) = a_P \left(\frac{1}{\nu_T} \frac{1}{\nu_R} \right)^{1/2}, \quad \hat{k}_P(t) = k_P \left(\nu_w \nu_f \nu_T \nu_R \frac{\hat{k}_h}{k_h} \frac{\hat{k}_A}{k_A} \frac{\hat{k}_c}{k_c} \right)^{1/7} \tag{6}$$

The coupling terms $\hat{k}_h/k_h$, $\hat{k}_A/k_A$, and $\hat{k}_c/k_c$ encode physiological dependencies: taller plants with more leaf area can support larger fruit, while larger tassels (more spikelets) may compete with ear development. [28], [29], [30], [31]. Table 4 in Section 2.7 details the actual values used for typical (unmodulated) growth rates and carrying capacities used in this study.

For crops exhibiting distinct vegetative and reproductive phases, double logistic models can capture two successive S-curves—rapid vegetative expansion followed by a second sigmoid for fruit or grain filling. However, since our optimization objective is end-of-season yield (and hence revenue), the precise shape of intermediate growth dynamics has limited impact on the optimal input strategy. We therefore retain the single logistic formulation as a reduced-order model that balances fidelity with computational tractability.

2.3. Delayed absorption via FIR convolution

Plants do not immediately process applied nutrients; instead, there is a physiologically-mediated delay between application and utilization. We model this delayed absorption using finite impulse response (FIR) convolution with Gaussian kernels.

If cumulative nutrient uptake follows a sigmoid trajectory—with slow initial uptake due to transport lag, rapid increase once metabolic pathways activate, and eventual saturation—then instantaneous nutrient absorption

follows a bell-shaped curve. A Gaussian kernel is the least assumptive choice for a bell curve, requiring only two parameters: the temporal spread σ (characterizing absorption duration) and the peak delay μ.

Given only the temporal spread σ for each nutrient type, we determine μ (hours after application where peak absorption occurs) such that 95% of the kernel mass lies within $[0, 2\mu]$. This requires solving

$$\operatorname{erf}\left(\frac{\mu}{\sigma\sqrt{2}}\right) = 0.95, \tag{7}$$

which yields $\mu \approx 1.96\sigma$. The Gaussian FIR kernel is then

$$G[k] = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\frac{(k-\mu)^2}{\sigma^2}\right\}. \tag{8}$$

for $k \geq 0$ hours.

The FIR horizon L^* is chosen as the minimum length capturing 95% of the kernel mass:

$$L^* = \min_L \left\{ L : \frac{\sum_{k=0}^{L-1} G[k]}{\sum_{k=0}^{K-1} G[k]} \geq 0.95 \right\} \tag{9}$$

where K is the simulation length in hours. Figure 2 panel 1 visualizes this transformation.

Absorbed nutrient signals are obtained from applied nutrients by convolving the applied signal with the corresponding Gaussian kernel. Different nutrients have different metabolic timescales, so we use $\sigma_w = 30$ hours for water (rapid uptake), $\sigma_f = 300$ hours for fertilizer (slow uptake reflecting root absorption dynamics), and $\sigma_T = \sigma_R = 30$ hours for temperature and radiation (immediate physiological effects with short memory). In future additions to the model, rate of water uptake from soil could be incorporated here.

2.4. Cumulative stress tracking via EMA filtering

While FIR convolution captures delayed absorption, plants also accumulate stress from sustained deviations from optimal conditions. Intuitively, a plant "expects" to receive nutrients at a steady rate throughout the season; prolonged under- or over-supply relative to this expectation causes stress. We formalize this by comparing cumulative absorbed nutrients to a linear

reference trajectory based on seasonal requirements from agronomic literature. To capture the delayed effects from the stress, we then additionally apply an exponential moving average (EMA) filters, which is equivalent to first-order infinite impulse response (IIR) filter.

The EMA filter with memory parameter $\beta \in [0, 1)$ has the recursive form:

$$y[k] = (1 - \beta)x[k] + \beta y[k - 1] \tag{10}$$

where larger β values correspond to longer memory (slower response to changes). This formulation preserves constant signals ($x[k] = c$ implies $y[k] \to c$) while smoothing transient fluctuations. In this study, we use $\beta_\Delta = 0.95$, the smoothing effects of which are demonstrated in Figure 2 panel 4.

2.5. Nutrient factor calculation

We now describe the complete transformation pipeline that converts raw input signals into nutrient factors $\nu \in [0, 1]$ that modulate plant growth. For a nutrient signal f (could represent water, fertilizer, temperature, or solar radiation):

Step 1: Delayed absorption. Convolve the raw signal $f[k]$ with the Gaussian FIR kernel:

$$\bar{f}[k] = \sum_{n=0}^{L-1} G[n] f[k - n] \tag{11}$$

Step 2: Cumulative absorption. Compute the running sum of absorbed nutrient:

$$F[k] = \sum_{n=0}^{k} \bar{f}[n] \tag{12}$$

Step 3: Instantaneous anomaly. Compare actual cumulative absorption to expected levels:

$$\delta[k] = \left| \frac{k \cdot f_{\text{typ}} - F[k]}{k \cdot f_{\text{typ}} + \epsilon} \right| \tag{13}$$

where f_{typ} is the typical hourly nutrient level the plant "expects" and ϵ is a small constant preventing division by zero.

Step 4: Cumulative divergence. Apply EMA smoothing to track sustained anomalies:

$$\Delta[k] = \beta_\Delta \Delta[k - 1] + (1 - \beta_\Delta)\delta[k] \tag{14}$$

where $\beta_\Delta = 0.95$ provides long memory.

Step 5: Nutrient factor. Convert divergence to a stress factor via exponential decay with additional EMA smoothing:

$$\nu[k] = \beta_\nu \nu_f[k-1] + (1-\beta_\nu)\exp\{-\alpha\Delta[k]\} \tag{15}$$

where $\alpha = 3$ ensures $\nu \approx 0.05$ when $\Delta = 1$ (complete divergence from expected levels), and $\beta_\nu = 0.05$.

The nutrient factor $\nu[k]$ equals 1 when nutrient application perfectly matches expected levels and decays toward 0 under sustained over- or under-application. This captures the intuition that plants are resilient to brief deviations but suffer cumulative damage from prolonged stress. Figure 2 illustrates the FIR convolution and EMA smoothing operations that constitute the metabolic transformation pipeline.

2.6. Simulation

The logistic ODEs (1) admit a closed-form solution, enabling exact time-stepping without numerical integration error. Given state $x(t)$ at time t, the state at $t + \Delta t$ is:

$$x(t + \Delta t) = \frac{\hat{k}_x(t)}{1 + \left(\frac{\hat{k}_x(t)}{x(t)} - 1 \right) \exp(-\hat{a}_x(t)\Delta t)} \tag{16}$$

where $\hat{a}_x(t)$ and $\hat{k}_x(t)$ are treated as constant over the time step. This closed-form approach is more accurate than forward Euler integration and avoids instability issues that can arise with explicit methods at larger time steps.

We simulate the growing season at hourly resolution ($\Delta t = 1$ hour), yielding approximately 2900 time steps for a typical corn season (late April to early October). At each step, we: (1) update the nutrient factors based on cumulative inputs and divergences, (2) compute effective growth rates and carrying capacities, and (3) advance each state variable using Equation 16.

2.7. Case study: corn in Iowa

We demonstrate the framework using corn, the most widely planted crop in the United States with over 90 million acres harvested annually [32]. The case study uses historical weather data from Fairfax, Iowa (41.76°N, 91.87°W), a representative location in the Corn Belt (USDA climate zones 4b–5b).

A single simulation covers a typical growing season from late April to early October (approximately 2900 hours). Environmental inputs are:

- **Temperature and radiation:** Hourly data from NSRDB for Fairfax, IA. Mean temperature is 22.8°C; mean solar radiation is 580 W/m^2 [33].
- **Precipitation:** Daily data from NOAA, interpolated to hourly resolution [34].
- **Typical nutrient expectations:** Based on agronomic recommendations [35], the model expects 28 inches of water and 355 lbs of NPK fertilizer over the season ($u_{w,\text{typ}} \approx 0.01$ in/hr, $u_{f,\text{typ}} \approx 0.12$ lb/hr).

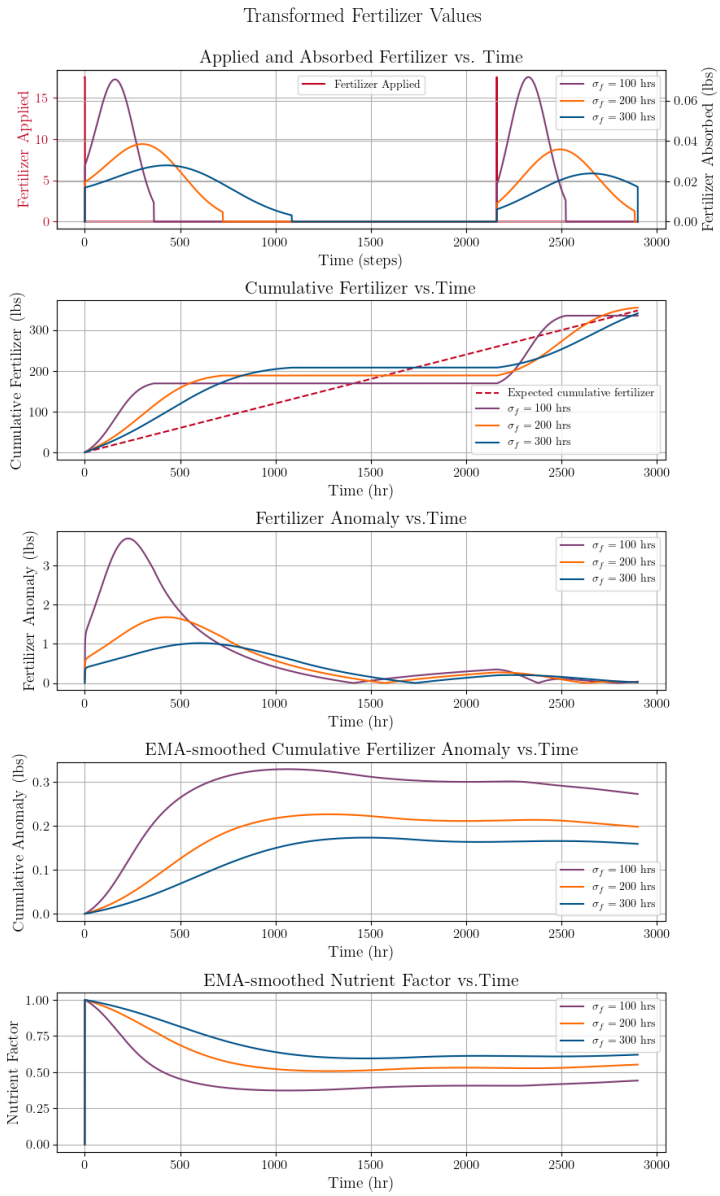

Figure 2: Illustration of the metabolic transformation pipeline using fertilizer as an example. Applied fertilizer is treated as a finite impulse, while absorbed fertilizer is modeled by convolving the applied signal with a Gaussian FIR kernel. Panel 1 shows how larger temporal spreads, which define the kernel, result in slower absorption. Panel 2 shows the cumulative fertilizer absorbed over time. Panel 3 shows the fertilizer anomaly, which compares the actual cumulative fertilizer absorbed to the expected level. Panel 4 shows how EMA filtering smooths the effects of cumulative anomalies. Panel 5 shows the nutrient factors that result from the preceding transformations.

State variable	Days to maturity	Hours to maturity	Typical final value
Plant height h	65–70	1560–1680	2.7–3.7 m
Leaf area A	55–65	1320–1560	0.6–0.7 m^2
Number of leaves N	65	1560	18–20
Spikelets c	65–70	1560–1680	∼1000
Fruit biomass P	125	3000	0.15–0.36 kg

Table 3: Corn development timeline and typical final values from agronomic literature. [36, 37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47].

State	Growth rate	Carrying capacity	Initial condition	Units
Height h	$a_h = 0.010$ hr^{-1}	$k_h = 3.0$	$h_0 = 0.001$	m
Leaf area A	$a_A = 0.0105$ hr^{-1}	$k_A = 0.65$	$A_0 = 0.001$	m^2
Leaves N	$a_N = 0.011$ hr^{-1}	$k_N = 20$	$N_0 = 0.001$	count
Spikelets c	$a_c = 0.010$ hr^{-1}	$k_c = 1000$	$c_0 = 0.001$	count
Fruit P	$a_P = 0.005$ hr^{-1}	$k_P = 0.25$	$P_0 = 0.001$	kg

Table 4: Model parameters derived from values in Table 3. Initial conditions are set to k_x/K where $K \approx 2900$ is the season length in hours (about 120 days).

Table 3 summarizes expected corn development timelines used to calibrate model parameters. The baseline growth rates and carrying capacities are crop-specific parameters that can be estimated from field data or literature values. For corn, we use the values in Table 4, calibrated to match typical development timelines where the corn reaches full vegetative size around 65–70 days after sowing and grain fill completes around 125 days [36]. For the GA cost function weights to be specific to corn, we use the values in Table 5, which were also taken from the literature.

3. Stochastic weather scenario generation

To evaluate optimizer robustness, we generate a suite of stochastic weather scenarios from historical data. Each scenario applies perturbations representing different climate conditions, from normal variability to extreme events. A single scenario can comprise multiple weather events. In this study, the possible types of weather events are droughts/downpours $\mathcal{D}$ and heat waves/-cold snaps $\mathcal{H}$. The process of adding perturbations to the historical baseline data in order to obtain synthetic, stochastic, and at times extreme, weather is as follows: 1) scale the precipitation and radiation data by fixed scaling

Parameter	Value	Derivation
w_w	\$2.00/inch	Typical irrigation cost
w_f	\$0.61/lb	Weighted NPK cost
w_h	\$35/m	Silage value proxy
w_A	\$215/m^2	Silage value proxy
w_P	\$4,450/kg	\$4/bushel $\times$ plant density

Table 5: Economic weights for the corn objective function. The fruit biomass weight accounts for approximately 28,350 plants per acre at \$0.157/kg. [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62].

factors, offset the temperature data by a fixed value, 2) add white noise to each time step, independent from other time steps, 3) inject all events of type $\mathcal{D}$, 4) inject all events of type $\mathcal{H}$, and 5) check that synthetic data falls within reasonable physical bounds. Details of those transformations follow below.

3.1. Defining a stochastic weather scenario

Let the historical hourly time series for the input disturbances be

$$d_S^{\text{hist}}[k] \geq 0 \tag{17}$$

$$-20 \leq d_T^{\text{hist}}[k] \leq 50 \tag{18}$$

$$d_R^{\text{hist}}[k] \geq 0 \tag{19}$$

for $k = 0, ..., K-1$ where K is the length of the growing season in hours, respectively representing precipitation (inches), temperature (°C), and solar radiation (W/m^2). Then, let a stochastic weather scenario be defined by the parameters

$$\mathcal{W} = (s_S, s_T, s_R, \eta, D, H) \tag{20}$$

where

- $s_S \in [0, 2]$ is the precipitation adjustment (scaling factor)
- $s_T \in \mathbb{R}$ is the temperature adjustment (offset)
- $s_R \in [0, 2]$ is the radiation adjustment (scaling factor)
- $\eta \in [0, 1]$ is the relative noise level (a fraction of the historical standard deviation)

- D is a set of drought or downpour events, $\mathcal{D}$
- H is a set of heatwave or cold snap events, $\mathcal{H}$

Each event type ($\mathcal{D}$ or $\mathcal{H}$) is defined by three parameters:

$$[k_0, \kappa, \iota] \tag{21}$$

where

- k_0 is the hour when the event begins
- κ is the duration of the event in number of hours
- $\iota \in [0, 1]$ is the intensity, with 1 being the most intense

3.1.1. Global scaling and offset

Initialize the synthetic environmental disturbance time series by applying the global adjustments (s_S, s_T, s_R) for $k = 0, ..., K-1$ as below

$$\begin{aligned} d_S^{(1)}[k] &= s_S d_S^{\text{hist}}[k] && \text{(scaling)} \\ d_T^{(1)}[k] &= d_T^{\text{hist}}[k] + s_T && \text{(offset)} \\ d_R^{(1)}[k] &= s_R d_R^{\text{hist}}[k] && \text{(scaling)} \end{aligned} \tag{22}$$

3.1.2. Add white noise

If $\eta > 0$, draw independent white noise sequences

$$\varepsilon_T[k] \sim \mathcal{N}(0, (\eta\sigma_T)^2) \tag{23}$$

$$\varepsilon_R[k] \sim \mathcal{N}(0, (\eta\sigma_R)^2) \tag{24}$$

where σ_T and σ_R are the standard deviations of the historical temperature and radiation time series, respectively.

We then smooth the white noise with a moving average over $m_T = 24$ hours for temperature and $m_R = 12$ hours for radiation, as we will only apply the noise to the radiation during the day time (when it is not as close to zero).

$$\bar{\varepsilon}_T[k] = \frac{1}{m_T} \sum_{n=0}^{m_T-1} \varepsilon_T[k-n] \tag{25}$$

$$\bar{\varepsilon}_R[k] = \frac{1}{m_R} \sum_{n=0}^{m_R-1} \varepsilon_R[k-n] \tag{26}$$

The smoothed noise is then added to the signals from section 3.1.1.

$$d_T^{(2)}[k] = d_T^{(1)}[k] + \bar{\varepsilon}_T[k] \tag{27}$$

$$d_R^{(2)}[k] = d_R^{(1)}[k] + \mathcal{M}_{\text{day}}\bar{\varepsilon}_R[k] \tag{28}$$

where

$$\mathcal{M}_{\text{day}} = \{k | d_R^{\text{hist}}[k] > d_R^{\text{day}}\} \tag{29}$$

and we choose the threshold $d_R^{\text{day}} = 10$ W/m^2. As our historical data for precipitation is daily, not hourly, precipitation remains unchanged in this step, so

$$d_S^{(2)}[k] = d_S^{(1)}[k] \tag{30}$$

If $\eta = 0$, then we let $(d_S^{(2)}, d_T^{(2)}, d_R^{(2)}) = (d_S^{(1)}, d_T^{(1)}, d_R^{(1)})$.

3.1.3. Drought/downpour injection

A drought or downpour event is

$$\mathcal{D} = [k_0, \kappa, \iota] \tag{31}$$

and drought events only affect synthetic precipitation. Each event takes place for a limited amount of time, so only some time steps are affected. For each event, we define the affected hourly index set as

$$\mathcal{I} = \{k_0, k_0 + 1, ..., \min(k_0 + \kappa - 1, K - 1)\}, \tag{32}$$

so either κ steps or the end of the season, whichever comes first, and then apply the intensity scaling to those indices with

$$d_S^{(3)}[k] \leftarrow (1 - \iota)d_S^{(2)}[k] \quad \text{for all } k \in \mathcal{I} \tag{33}$$

Temperature and radiation are unchanged in this step, so

$$d_T^{(3)}[k] = d_T^{(2)}[k] \quad \text{and} \quad d_R^{(3)}[k] = d_R^{(2)}[k] \tag{34}$$

3.1.4. Heat wave/cold snap injection

A heat wave or cold snap event affects only synthetic temperature and is defined by

$$\mathcal{H} = [k_0, \kappa, \iota] \tag{35}$$

where k_0 and κ have the same meanings they did for the drought event, but now ι is the peak/valley temperature difference (an offset rather than a

scaling factor). We then construct a ramp-up, hold, and ramp-down for the event. Let

$$\kappa_{\text{ramp}} = \max(1, 0.1\kappa) \tag{36}$$

i.e. either one tenth of the heat wave duration or at least one hour. Then, let

$$\kappa_{\text{hold}} = \kappa - 2\kappa_{\text{ramp}} \tag{37}$$

to account for the ramp-up and ramp-down.

We can then define

- ramp-up: $w_{\text{up}}[k] = \dfrac{k}{\kappa_{\text{ramp}} - 1}$ for $k = 0, ..., \kappa_{\text{ramp}} - 1$
- ramp-hold: $w_{\text{hold}}[k] = 1$
- ramp-down: $w_{\text{down}}[k] = \dfrac{k}{\kappa_{\text{ramp}} - 1}$ for $k = 0, ..., \kappa_{\text{ramp}} - 1$

and concatenate to form the heat wave window

$$w_{\text{heat}} = [w_{\text{up}}, w_{\text{hold}}, w_{\text{down}}] \tag{38}$$

Applying the event to the temperature time series, we obtain

$$d_T^{(4)}[k] \leftarrow d_T^{(3)}[k] \pm \iota w_{\text{heat}}[k - k_0] \tag{39}$$

for $k = k_0, k_0 + 1, ..., \min(k_0 + \kappa - 1, K - 1)$, where ι is added in the case of a heat wave and subtracted in the case of a cold snap. Precipitation and radiation are unchanged in this step, so

$$d_S^{(4)}[k] = d_S^{(3)}[k] \quad \text{and} \quad d_R^{(4)}[k] = d_R^{(3)}[k] \tag{40}$$

3.1.5. Clipping to physical bounds

Finally, we check to ensure that the transformations above have not violated the bounds on the input disturbances specified in equations 19. If they have, we clip the values as below

$$d_S^{\text{syn}}[k] = \max(0, d_S^{(4)}[k]) \tag{41}$$

$$d_T^{\text{syn}}[k] = \min(50, \max(-20, d_T^{(4)}[k])) \tag{42}$$

$$d_R^{\text{syn}}[k] = \max(0, d_R^{(4)}[k]) \tag{43}$$

for all $k \in \{0, ..., K - 1\}$.

3.2. Extremity index

To systematically compare strategy performance across diverse conditions, we define an extremity index $\mathcal{E}$ that quantifies how far a scenario deviates from baseline conditions. This index serves two purposes: (1) analyzing whether different optimization methods perform better at different extremity levels, and (2) ensuring that MPC parameter tuning covers a representative range of conditions. The index aggregates contributions from precipitation anomalies, temperature anomalies, and discrete weather events, with each component scaled so that a value of 1 represents a meaningful agronomic threshold—i.e., a deviation large enough to require management intervention or cause measurable yield impact.

Precipitation extremity. The precipitation scaling factor s_S determines seasonal water availability, with $s_S = 1$ representing baseline conditions. Deviations in either direction—drought ($s_S < 1$) or downpour ($s_S > 1$)—contribute to extremity:

$$\mathcal{E}_{\text{precip}} = 2|1 - s_S| \tag{44}$$

The factor of 2 is chosen so that a 50% reduction in precipitation ($s_S = 0.5$) yields $\mathcal{E}_{\text{precip}} = 1$. This threshold reflects agronomic practice: a 50% precipitation deficit during the growing season typically triggers supplemental irrigation and is considered a moderate drought requiring active management [63].

Temperature extremity. Seasonal temperature offsets contribute to extremity based on their magnitude:

$$\mathcal{E}_{\text{temp}} = \frac{|s_T|}{5} \tag{45}$$

The ±5°C threshold reflects corn physiology: sustained temperatures above 35°C or below 10°C significantly reduce photosynthesis and pollen viability [64]. Since mean baseline temperature is approximately 23°C, a 5°C offset moves conditions into ranges where yield impacts become measurable.

Drought event extremity. Discrete drought events contribute based on their duration and intensity:

$$\mathcal{E}_{\text{drought}} = \sum_{(k_0,\kappa,\iota)\in D} \frac{\kappa}{500}\iota \tag{46}$$

The 500-hour (approximately 21-day) normalization reflects the timescale over which soil moisture depletion becomes critical. Corn can typically tolerate 7–10 days without rainfall before showing stress symptoms; by three weeks, yield losses become significant without irrigation intervention [63]. A full-intensity drought lasting 21 days thus represents the threshold for serious crop impact.

Heat wave and cold snap extremity. Temperature events contribute based on both duration and intensity:

$$\mathcal{E}_{\text{heat}} = \sum_{(k_0,\kappa,\iota)\in H} \frac{\kappa}{200}\frac{\iota}{5} \tag{47}$$

The 200-hour (approximately 8-day) duration threshold reflects the timescale over which sustained heat stress affects corn reproductive development. During tasseling and silking, even brief heat waves can reduce kernel set, but cumulative effects become severe after about one week [64]. The 5°C intensity threshold again corresponds to the deviation that moves temperatures into physiologically stressful ranges.

Aggregate extremity. The total extremity index is the sum of all components:

$$\mathcal{E} = \mathcal{E}_{\text{precip}} + \mathcal{E}_{\text{temp}} + \mathcal{E}_{\text{drought}} + \mathcal{E}_{\text{heat}} \tag{48}$$

3.3. Weather scenario suite

We evaluate strategies across 21 stochastic weather scenarios generated from the baseline historical data. Table 6 summarizes the scenario categories.

Category	Count	Extremity Range
Normal	5	0.03–0.18
Moderate	5	0.35–0.90
Drought	4	1.20–2.45
Wet/Cool	2	0.95–1.10
Heat Stress	2	2.10–3.50
Extreme	2	5.60–8.38

Table 6: Weather scenario categories with extremity index ranges.

4. Genetic algorithm optimization

Given the nonlinear, delay-affected dynamics of the crop model, gradient-based optimization is challenging. The delayed effects of inputs create a non-convex landscape with potentially many local optima. We therefore employ a genetic algorithm (GA), a population-based metaheuristic inspired by natural selection, that can effectively explore complex search spaces [14].

4.1. Algorithm description

The GA maintains a population of M candidate solutions and iteratively improves them through selection, crossover, and mutation over G generations. Figure 3 presents the complete procedure.

Selection. After each generation, population members are ranked by cost. In our application, we use negative revenue as the cost, as this is a minimization problem. The top P members survive as "parents" for the next generation. This selection ensures the best solutions are never lost.

Crossover. New "children" C are created by blending two parent solutions. For each child, we randomly select two parents and compute a weighted average:

$$\mathbf{u}^{(\text{child})} = \phi \cdot \mathbf{u}^{(a)} + (1 - \phi) \cdot \mathbf{u}^{(b)} \tag{49}$$

where $\phi \sim \text{Uniform}(0, 1)$ under normal operation. This crossover can produce children anywhere along the line segment connecting the parents, enabling smooth exploration of the search space.

Mutation and Diversity. To maintain population diversity and escape local optima, the remaining $M - P - C$ population slots are filled with randomly generated solutions. Additionally, if the best cost stagnates (changes by less than \$0.01) for 10 consecutive generations, we switch to aggressive crossover with $\phi \sim \text{Uniform}(-1, 2)$. This allows children to lie outside the convex hull of their parents, promoting exploration of new regions.

Default Parameters. We use $M = 128$ members, $P = 16$ parents, $C = 16$ children, and $G = 100$ generations. The large population relative to generations ensures diversity for exploration while preventing premature convergence.

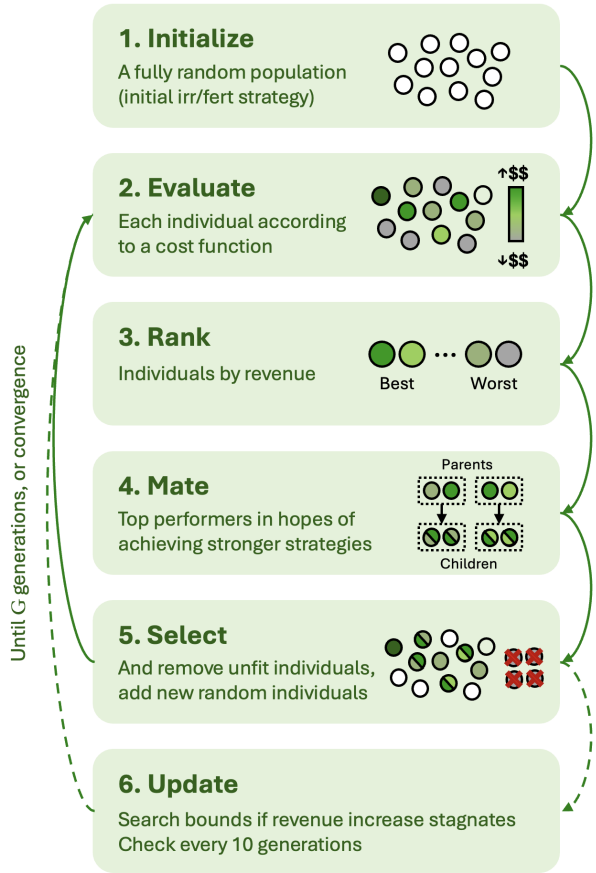

Figure 3: Genetic algorithm visualized. Initialization, followed by evaluation, ranking, mating, and selection, repeated for 100 generations with an added check for stagnation every 10 generations.

4.2. Decision variables and search space: GA

Each candidate solution encodes a complete irrigation and fertilization strategy as a four-dimensional vector:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} \text{irrigation frequency (hours)} \\ \text{irrigation amount (inches)} \\ \text{fertilizer frequency (hours)} \\ \text{fertilizer amount (lbs)} \end{bmatrix} \tag{50}$$

The frequencies specify application intervals: $u_1 = 168$ means irrigate every 168 hours (weekly). The amounts specify the quantity applied at

each event. This parameterization assumes regular, periodic application—a simplification that captures common agricultural practice while keeping the search space tractable.

The GA searches over the following bounds:

- Irrigation frequency: 100–700 hours (4–29 days between applications)
- Irrigation amount: 0.5–5.0 inches per application
- Fertilizer frequency: 700–2900 hours (29–121 days, i.e., 1–4 applications per season)
- Fertilizer amount: 100–500 lbs per application

4.3. Genetic algorithm objective function

The objective is to maximize net revenue, defined as crop value minus input costs.

$$\text{Revenue}(\mathbf{u}) = \text{Crop Value}(\mathbf{x}) - \text{Input Costs}(\mathbf{u}) \tag{51}$$

The crop value depends on final plant state at harvest:

$$\text{Crop Value}(\mathbf{x}) = w_h x_h[K] + w_A x_A[K] + w_P x_P[K] \tag{52}$$

where K is the final time step and w_h, w_A, w_P are economic weights (dollars per unit) for height, leaf area, and fruit biomass respectively.

The input costs accumulate over the season:

$$\text{Input Costs}(\mathbf{u}) = w_w \sum_{k=0}^{K} u_2[k] + w_f \sum_{k=0}^{K} u_4[k] \tag{53}$$

where w_w and w_f are costs per unit of irrigation and fertilizer.

For corn, the economic weights are derived from market prices and typical yields (Table 5). The fruit biomass weight dominates, reflecting that grain yield is the primary economic output.

5. Model predictive control

5.1. Optimization and decision epochs

Model Predictive Control (MPC) frames resource allocation as a sequence of optimization problems. At each *decision epoch*—in our case, the start of each day—the controller observes the current system state and solves an optimization problem to determine the best actions over a finite planning horizon. The solution specifies how much irrigation and fertilizer to apply, balancing immediate costs against long-term crop development.

This approach differs fundamentally from the genetic algorithm method, which optimizes a single fixed strategy over the entire growing season. MPC instead solves a smaller optimization problem at each decision epoch, using updated state and forecast information to adapt to changing conditions.

5.2. Constrained finite-time optimal control

At each decision epoch, model predictive control solves a constrained finite-time optimal control (CFTOC) problem over a planning horizon of K_d days. Discrete-time optimal control is concerned with choosing an optimal input sequence over the horizon K_d

$$\mathcal{U}_{0\to K_d} = \{\mathbf{u}[k]\} \quad \text{for} \quad k = 0, ..., K_d - 1 \tag{54}$$

with respect to some objective function over a finite or infinite time horizon in order to apply it to a system with a given initial state $\mathbf{x}[0]$. The objective function is often defined as a sum of stage costs $q(\mathbf{x}[k], \mathbf{u}[k])$ and when the horizon has finite length, a terminal cost $p(\mathbf{x}[K_d])$. That is

$$J_{0\to K_d}(\mathbf{x}[0], \mathcal{U}_{0\to K_d}) = p(\mathbf{x}[K_d]) + \sum_{n=0}^{K_d-1} q(\mathbf{x}[n], \mathbf{u}[n]) \tag{55}$$

where the states $\mathcal{X}_{0\to K_d} = \{\mathbf{x}[k]\} \quad \text{for} \quad k = 0, ..., K_d - 1$ must satisfy the initial condition and system dynamics

$$\begin{cases} \mathbf{x}[0] = \mathbf{x}_0 \\ \mathbf{x}[k+1] = g(\mathbf{x}[k], \mathbf{u}[k]) \quad \text{for } k = 0, ..., K_d - 1 \end{cases} \tag{56}$$

and there may be other state or input constraints formulated as inequalities

$$h(\mathbf{x}[k], \mathbf{u}[k]) \leq 0 \quad \text{for} \quad k = 0, ..., K_d - 1 \tag{57}$$

In the finite horizon case, there may also be a terminal constraint requiring the final state to lie in some terminal set $\mathbf{x}[K] \in \mathcal{X}_{\text{final}}$.

In our specific case, we construct the problem as a minimization

$$J^*_{0\to K_d}(\mathbf{x}[0]) = \min_{\mathcal{U}_{0\to K_d}} J_{0\to K_d}(\mathbf{x}[0], \mathcal{U}_{0\to K_d}) \quad \text{s.t.}$$
$$\begin{cases} \mathbf{x}[0] = \mathbf{x}_0 \\ \mathbf{x}[k+1] = g(\mathbf{x}[k], \mathbf{u}[k]) \quad \text{for } k = 0, ..., K_d - 1 \\ \mathbf{x} \in \mathbb{R}^+ \\ \mathbf{u} \in \mathcal{U} \end{cases} \tag{58}$$

where the stage cost penalizes input usage and nutrient anomalies:

$$q(\mathbf{x}[k], \mathbf{u}[k]) = \underbrace{\omega_w \left(\frac{u_w[k]}{u_{w,\text{typ}}}\right)^2 + \omega_f \left(\frac{u_f[k]}{u_{f,\text{typ}}}\right)^2}_{\text{input usage}} + \underbrace{\omega_{\Delta w}(\Delta_w[k])^2 + \omega_{\Delta f}(\Delta_f[k])^2}_{\text{nutrient anomalies}} \tag{59}$$

Here ω_w and ω_f are weights on normalized irrigation and fertilizer inputs, $u_{w,\text{typ}}$ and $u_{f,\text{typ}}$ are typical application rates, and $\omega_{\Delta w}$ and $\omega_{\Delta f}$ penalize the cumulative nutrient anomalies Δ_w and Δ_f defined in (14). The terms in the stage cost have been squared in order to encourage sparsity in actuation, more similar to what a farmer might actually implement.

The terminal cost rewards crop development:

$$p(\mathbf{x}[K_d]) = -\omega_h \frac{x_h[K_d]}{k_h} - \omega_A \frac{x_A[K_d]}{k_A} - \omega_P \frac{x_P[K_d]}{k_P} \tag{60}$$

where ω_h, ω_A, and ω_P weight the normalized final height, leaf area, and fruit biomass, respectively, and we treat a reward as a negative cost. There is no terminal set constraint because we want the plant to grow as much as possible.

5.3. Decision variables and search space: MPC

The MPC controller uses a 9-day planning horizon ($K_d = 9$) with daily re-optimization. Control bounds are set to $u_w \in [0, 0.7]$ inches/hour and $u_f \in [0, 12]$ lbs/hour, reflecting practical application rate limits.

The CFTOC problem is formulated in Pyomo [65] and solved using the IPOPT interior-point nonlinear optimizer [66] with the MUMPS linear solver. The nonlinearity arises from the logistic dynamics, FIR convolution, EMA filtering, and exponential nutrient factor computation.

5.4. MPC parameter tuning via Bayesian optimization

The MPC cost function contains seven tunable weights: $\omega_w, \omega_f, \omega_{\Delta w}, \omega_{\Delta f}, \omega_h, \omega_A, \omega_P$, plus the horizon length K_d. These parameters significantly affect controller behavior, and their optimal values depend on the weather scenario ensemble. We use Bayesian Optimization (BO) to tune these parameters for robust performance.

Bayesian Optimization (BO) is well-suited to hyperparameter tuning problems where function evaluations are expensive and gradients are unavailable [67]. The approach maintains a probabilistic model of the objective function and uses this model to decide where to sample next, balancing exploration of uncertain regions against exploitation of promising regions.

Tree-structured Parzen Estimation (TPE) [68] models the posterior (probability of particular MPC parameter set, given a particular revenue), rather than the likelihood (revenue, given a parameter set), using two density functions $\ell(x)$ and $g(x)$. Observations are partitioned by a quantile threshold γ (typically 0.1–0.25) into "good" and "bad" sets, with corresponding densities $\ell(x) = p(x|y \leq y^-)$ and $g(x) = p(x|y > y^-)$, where y^- is the γ-quantile of observed values. Starting from expected improvement and applying Bayes' rule, one can show that maximizing expected improvement is equivalent to maximizing $\ell(x)/g(x)$, which serves as the TPE acquisition function.

TPE approximates the densities $\ell(x)$ and $g(x)$ using kernel density estimation (KDE), placing Gaussian kernels around observed data points. For multivariate parameters, TPE assumes independence across dimensions, and the "tree-structured" aspect accommodates conditional parameter dependencies. TPE offers computational advantages over Gaussian-process-based methods and handles mixed parameter spaces (continuous, discrete, categorical) making it well-suited for MPC parameter tuning where the objective landscape may contain discontinuities from solver failures and the search space includes both continuous weights and integer horizon length.

Table 7 defines the search space for MPC parameter tuning. Stage cost weights use log-scale sampling to span multiple orders of magnitude, while terminal cost weights use linear sampling (as ask prices for crops do not fluctuate over orders of magnitude). Integer sampling is used for the horizon.

For robust parameter tuning, we optimize average performance across multiple weather scenarios:

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \frac{1}{|\mathcal{S}|} \sum_{\mathcal{S}_i \in \mathcal{S}} \text{Revenue}(\text{MPC}(\boldsymbol{\theta}, \mathcal{S}_i)) \tag{61}$$

Parameter	Lower	Upper	Scale	Optimal
ω_w (irrigation cost)	0.001	10.0	log	1.43
ω_f (fertilizer cost)	0.0001	1.0	log	4.81×10^{-3}
$\omega_{\Delta w}$ (water anomaly)	0.001	10.0	log	1.88
$\omega_{\Delta f}$ (fertilizer anomaly)	0.001	10.0	log	0.450
ω_h (height value)	10.0	1000.0	linear	323
ω_A (leaf area value)	10.0	1000.0	linear	629
ω_P (fruit biomass value)	100.0	10000.0	linear	1.95×10^3
K_d (horizon, days)	3	14	integer	9

Table 7: Bayesian optimization search space for MPC parameters.

where $\mathcal{S}$ is a representative subset of 5 weather scenarios spanning the extremity range. We use $N = 100$ trials with $N_0 = 20$ random startup trials before engaging the TPE sampler. The optimal MPC parameter values resulting from the Bayesian optimization procedure are also included in Table 7.

5.5. Receding-horizon algorithm

MPC implements the CFTOC in a receding-horizon fashion, re-optimizing daily based on updated state and weather observations. At each decision day k, it

1. Observes current plant state $\mathbf{x}[k]$ and obtain K_d-day weather forecast.
2. Solves the CFTOC problem (58) to obtain optimal control sequence.
3. Applies only the first control $\mathbf{u}^*[k] = (u_w[k], u_f[k])$ uniformly over day k.
4. Simulates hourly plant dynamics over the day using actual (not forecast) weather.
5. Advances to day $k + 1$ and repeat.

The key advantage of MPC over open-loop optimization (such as GA) is its closed-loop nature. By re-solving the optimization problem each day with updated state and weather information, MPC can:

- **Correct for forecast errors:** If yesterday's weather differed from the forecast, today's optimization accounts for the actual plant state.
- **Adapt to changing conditions:** If a heat wave arrives unexpectedly, MPC can adjust irrigation to maintain nutrient factors.

Algorithm 1 Model Predictive Control for Irrigation and Fertilization

Input: Initial state $x[0]$, horizon K_d, season length K days, weather data
Output: Control history $\{u[k]\}_{k=0}^{K-1}$, final state $x[K]$

Initialize FIR kernel buffers, EMA state variables
for $k = 0$ to $K - 1$ **do**
    Aggregate hourly weather forecast into daily averages for days $k, \ldots, k + K_d - 1$
    Solve CFTOC$(x[k], \{(\hat{d}_S, \hat{d}_T, \hat{d}_R)_{k+i}\}_{i=0}^{K_d-1}) \rightarrow \{u^*[k+i]\}_{i=0}^{K_d-1}$
    Extract first control: $(u_w[k], u_f[k]) \leftarrow u^*[k]$
    for $t = 0$ to 23 **do** ▷ Hourly simulation within day k
        Apply $(u_w[k]/24, u_f[k]/24)$ with actual weather $(d_{S,24k+t}, d_{T,24k+t}, d_{R,24k+t})$
        Update FIR buffers, EMA states, nutrient factors
        Advance plant state using closed-form logistic solution (16)
    end for
    $x_{k+1} \leftarrow$ current plant state
end for
return $\{u[k]\}_{k=0}^{K-1}$, $x[K_d]$

- **Exploit updated forecasts:** Longer-range forecasts become more accurate as the event approaches.

6. Results

6.1. Simulation configuration

All simulations use a 121-day growing season representative of corn production in Iowa, with hourly resolution for plant dynamics and daily resolution for closed-loop control decisions. We generate 21 weather scenarios spanning normal conditions to extreme drought and heat stress using the stochastic weather generation procedure described in Section 3.

The crop model parameters are calibrated to literature values with carrying capacities of 3.0 m for plant height, 0.65 m^2 for leaf area, 20 leaves, 1000 flower spikelets, and 0.25 kg for fruit (ear) biomass. Absorption kernels use $\sigma_w = 30$ hr for water and $\sigma_f = 300$ hr for fertilizer, reflecting faster water uptake relative to slower fertilizer absorption. Economic parameters follow current USDA estimates: corn price of \$4/bushel, yield conversion of 200 bushels/acre at maximum biomass, water cost of \$2/acre-inch, and NPK fertilizer cost of \$0.61/lb. Net revenue is computed as gross revenue minus input costs.

6.2. Baseline performance: farmer best practices

To establish a performance baseline, we simulate crop growth using conventional farmer practices: weekly irrigation at 1 inch per application [63] and monthly fertilization at 90 lbs per application [69]. These values reflect standard agronomic recommendations for corn in the Corn Belt region [70], [71], [72].

Figure 4 shows the environmental disturbances and control inputs over the growing season under historical ("normal") conditions, as well as the periodic irrigation and fertilizer applications from farmer best practices.

Figure 5 shows the resulting plant state trajectories. Under normal conditions with conventional management, the plant reaches suboptimal final values: height of 2.6 m (vs. 3.0 m capacity), leaf area of 0.45 m^2 (vs. 0.65 m^2), and fruit biomass of 0.16 kg (vs. 0.25 kg). The net revenue under this scenario is \$587/acre. The baseline scenario demonstrates how the model captures cumulative stress effects: despite regular irrigation, the mismatch between applied water and the plant's metabolic expectations under drought conditions leads to sustained nutrient factor depression and reduced growth

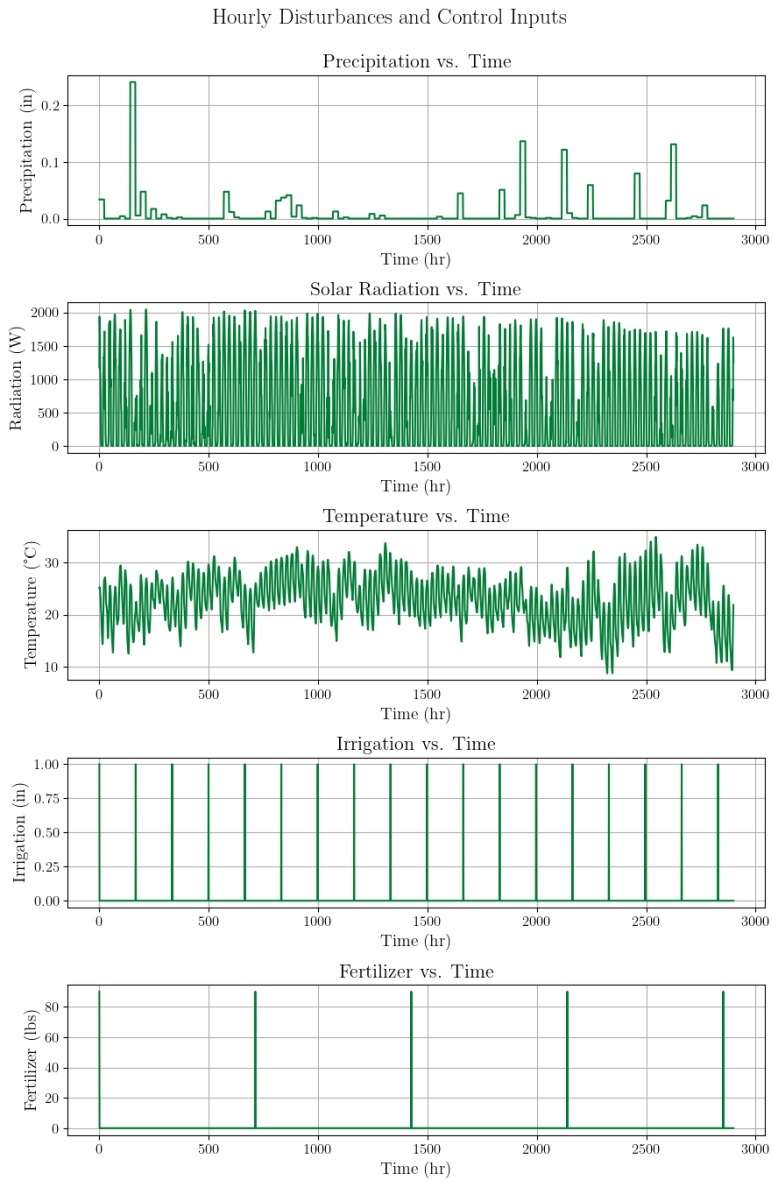

Figure 4: Environmental disturbances and control inputs for the baseline ("normal") scenario. Top three panels show hourly precipitation, solar radiation, and temperature from historical Iowa data. Bottom two panels show the farmer's irrigation (weekly, 1 inch) and fertilizer (monthly, 90 lbs) application strategy.

potential. Supplementary Figures A.11–A.14 provide detailed visualizations of the nutrient absorption dynamics and stress factor evolution.

For reference, current Iowa corn yields average approximately 200 bushels per acre [73] at prices near $4 per bushel [74]. Variable costs for irrigation and fertilizer are approximately $220 per acre [75], yielding estimated net revenue of roughly $580/acre under favorable conditions. Our simulated farmer best practices achieves $587/acre, suggesting reasonable model calibration.

6.3. Genetic algorithm optimization

We optimize irrigation and fertilization strategies using a genetic algorithm with 100 generations and population size of 128. Each individual encodes four decision variables: irrigation frequency (hours), irrigation amount (inches), fertilizer frequency (hours), and fertilizer amount (lbs). Fitness is evaluated as net revenue over the 121-day growing season.

To assess robustness, we executed 10 independent GA runs with different random seeds under normal weather conditions. Figure 6 shows the convergence of all 10 runs, with each curve representing the best revenue achieved at each generation. All runs converge to similar final values (within 3% of each other), demonstrating that the GA reliably finds near-optimal solutions despite the stochastic nature of the search.

The best GA strategy under normal conditions achieves $915/acre net revenue compared to $587/acre for farmer best practices—a 56% improvement. Figure 7 shows the plant state trajectories for the best member from each of the 10 independent GA runs. All optimized strategies achieve similar, near-optimal growth trajectories, confirming that different GA runs converge to similar optimal strategies.

To verify that the model appropriately discriminates between effective and ineffective strategies, we examined the plant state trajectories for the *worst* member in each GA run's final population (Supplementary Figure A.15). These strategies—the least fit survivors after 100 generations of evolution—achieve almost no growth and underperform the farmer best practices. This confirms that: (1) the GA successfully eliminates poor strategies through evolutionary pressure, and (2) arbitrary irrigation and fertilization schedules are indeed worse than conventional farmer practices, confirming the model is appropriately sensitive to irrigation and fertilizer input.

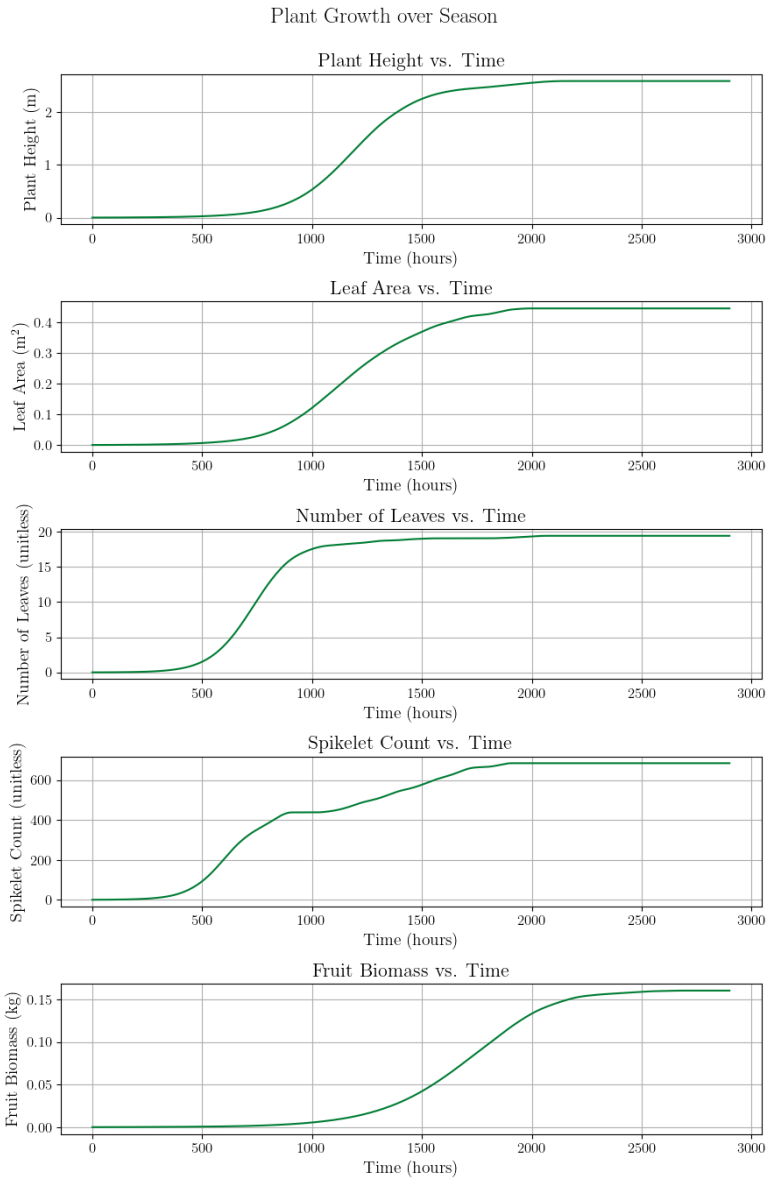

Figure 5: Plant state variable trajectories under the baseline scenario (farmer best practices during mild drought). All state variables reach suboptimal final values due to cumulative water stress. This strategy yields $587/acre in revenue.

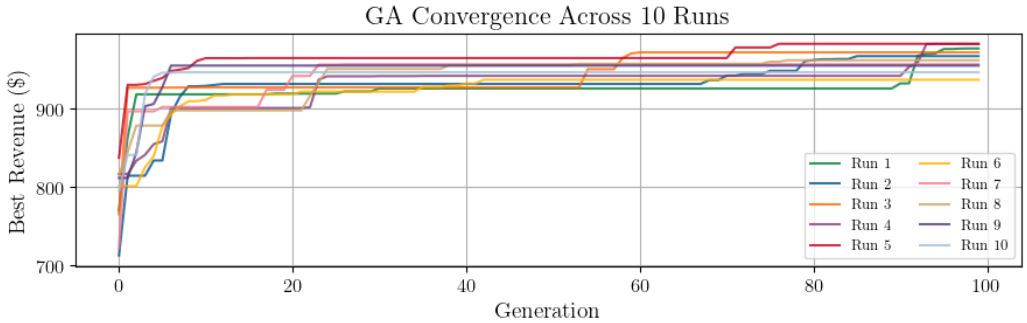

Figure 6: GA convergence across 10 independent runs. Each curve shows the best revenue at each generation. All runs exhibit rapid improvement in early generations followed by convergence to near-optimal solutions. The consistency across runs demonstrates algorithm robustness.

6.4. Four-way strategy comparison

We compare four irrigation and fertilization strategies across all 21 weather scenarios:

Farmer Best Practices. Standard agronomic recommendations: weekly irrigation at 1 inch per application and monthly fertilization at 90 lbs per application.

GA (Drought). A strategy optimized via genetic algorithm for drought conditions (50% of historical precipitation). The optimal strategy applies irrigation every 52 days at 5.0 inches per application, and fertilizer every 33 days at 77 lbs per application.

GA (Normal). A strategy optimized via genetic algorithm for normal weather conditions. The optimal strategy applies irrigation every 81 days at 2.1 inches per application, and fertilizer every 9 days at 30 lbs per application.

Adaptive MPC. Model Predictive Control that re-optimizes daily based on observed weather and current plant state, using a 9-day planning horizon as determined by Bayesian optimization.

Table 8 presents summary statistics for all four strategies. Figure 8 visualizes the statistics from Table 8. Choosing a strategy with less variance in revenue across weather scenarios could be considered less risk averse, in which case GA (Normal) would be the ideal choice.

All three optimization-based methods outperform farmer best practices, with mean advantages of $170/acre for GA (Drought), $215/acre for GA (Normal), and $123/acre for MPC. Figure 9 visualizes revenue across all 21

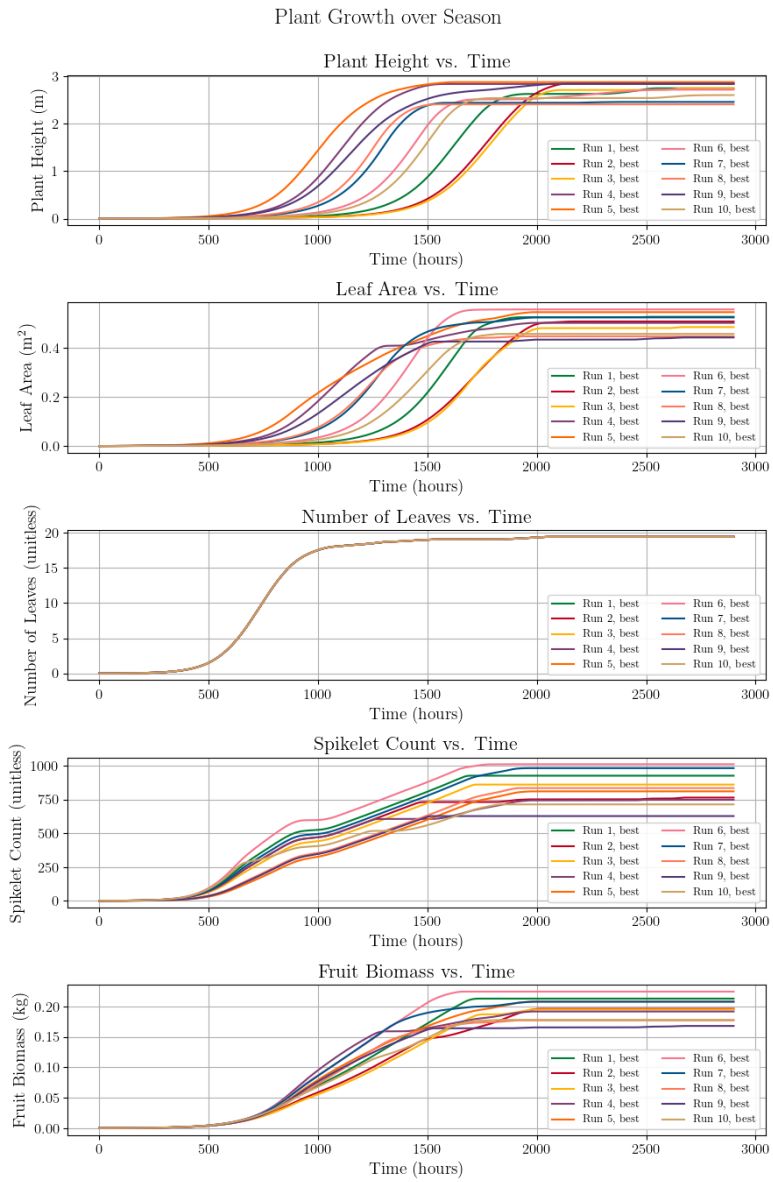

Figure 7: Plant state variable trajectories for the best member from each of the 10 independent GA runs. The tight clustering of trajectories demonstrates that different GA runs converge to similar optimal strategies.

Strategy	Mean ($/acre)	Std Dev	CV (%)	Min	Max
Farmer Best Practices	626	135	21.6	385	853
GA (Drought)	796	132	16.6	550	997
GA (Normal)	842	124	14.7	527	973
Adaptive MPC	750	115	15.4	445	886

Table 8: Summary statistics for the four strategies across 21 weather scenarios. CV = coefficient of variation (standard deviation / mean).

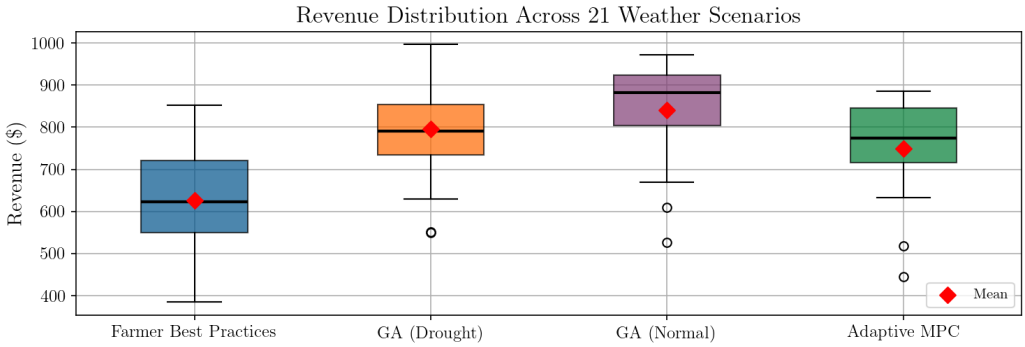

Figure 8: Revenue distribution across all 21 weather scenarios. Box plots show median, interquartile range, and outliers for each strategy. GA (Normal) achieves the highest median revenue but also exhibits the widest range, while MPC shows the lowest variability.

scenarios sorted by increasing extremity index.

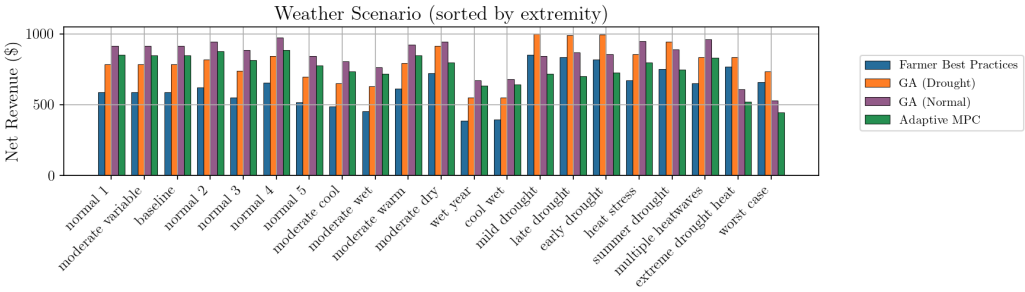

Figure 9: Revenue comparison across all 21 weather scenarios, sorted by extremity index. All optimization-based strategies outperform farmer best practices in most scenarios, with GA (Normal) achieving the highest revenues under normal conditions and GA (Drought) excelling under extreme conditions.

Figure 10 shows revenue comparison grouped by weather category. GA

(Normal) dominates under normal and moderate conditions, while GA (Drought) provides better performance under drought and extreme scenarios. MPC provides consistent but not optimal performance across all categories.

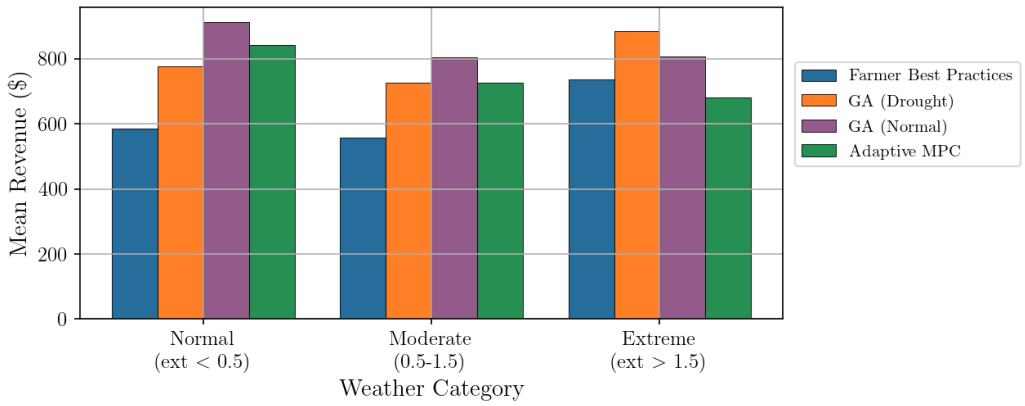

Figure 10: Revenue comparison by weather category. GA (Normal) achieves the highest revenues under normal conditions, while GA (Drought) is better suited for extreme weather scenarios.

Table 9 presents detailed results for all 21 scenarios. GA (Drought) wins all 21 scenarios against farmer best practices, GA (Normal) wins 18 of 21, and MPC wins 15 of 21. When comparing optimization methods directly, GA (Normal) outperforms MPC in all 21 scenarios, GA (Drought) outperforms MPC in 9 of 21 scenarios, and GA (Normal) outperforms GA (Drought) in 15 of 21 scenarios.

7. Discussion

7.1. Interpretation of results

The GA-optimized strategies differ from conventional wisdom in several notable ways. The algorithm discovers that less frequent but larger irrigation events, combined with reduced total fertilizer input, can outperform conventional uniform application schedules. This counterintuitive result emerges from the model's delayed absorption dynamics: the plant's metabolic expectations are calibrated to typical nutrient availability over time, and the GA discovers that strategically-timed resource inputs better maintain alignment with these metabolic expectations than aggressive compensation through frequent, uniform applications.

Scenario	Extremity	Farmer	GA (Drought)	GA (Normal)	MPC
normal 1	0.00	587	784	914	850
moderate variable	0.00	586	784	913	847
baseline	0.00	587	784	915	848
normal 2	0.20	622	819	945	877
normal 3	0.20	550	739	883	815
normal 4	0.40	652	842	973	886
normal 5	0.40	515	696	845	774
moderate cool	0.60	485	652	804	736
moderate wet	0.70	453	629	762	716
moderate warm	0.70	614	791	925	846
moderate dry	0.80	721	913	945	796
wet year	1.10	385	551	669	633
cool wet	1.10	393	550	680	641
mild drought	1.30	853	997	843	717
late drought	1.52	833	991	869	700
early drought	1.64	817	993	855	727
heat stress	1.70	669	855	948	795
summer drought	1.70	752	944	888	746
multiple heatwaves	2.26	651	837	959	829
extreme drought heat	4.97	768	836	609	518
worst case	8.38	659	734	527	445

Table 9: Revenue ($/acre) for all 21 weather scenarios sorted by extremity index. Cell shading indicates revenue level (darker green = higher revenue). GA (Normal) dominates in normal/moderate conditions; GA (Drought) excels in drought scenarios; all strategies struggle in extreme combined scenarios.

Under delayed absorption dynamics, timing matters more than total quantity. The FIR convolution introduces a temporal spread of 30 hours for water and 300 hours for fertilizer, meaning that today's application affects plant growth over the coming days to weeks. The GA can exploit this by front-loading or back-loading resources to align cumulative absorption with plant expectations throughout the season. Farmer best practices, with their rigid weekly/monthly schedules, cannot achieve this alignment.

MPC does not outperform the GA strategies, contrary to our initial hypothesis. The GA strategies, despite being "fixed," are optimized for their target conditions over the entire 121-day growing season. MPC re-optimizes daily but only sees 9 days ahead—insufficient foresight given the slow physi-

ological dynamics. With fertilizer absorption spreading over 300 hours (12+ days), MPC's horizon barely captures a single absorption cycle. This limited foresight prevents MPC from making the long-term resource allocation decisions that the GA can pre-compute. In agricultural systems with slow dynamics—where nutrient absorption takes days to weeks—the GA's full-season optimization horizon appears more valuable than MPC's adaptive but myopic decision-making.

Each GA strategy performs best when conditions match its optimization assumptions. GA (Normal) outperforms GA (Drought) in 12 of 21 scenarios, while GA (Drought) wins in the 9 scenarios with more extreme conditions. This is unsurprising: each strategy was optimized for its target conditions. The practical implication is that farmers should select the appropriate GA strategy based on seasonal forecasts—if drought is anticipated, use GA (Drought); otherwise, use GA (Normal).

Despite lower peak revenue, MPC offers a sustainability advantage worth noting. MPC achieves competitive returns (outperforming farmer best practices in 15 of 21 scenarios) while using dramatically less water (1.3–2.0 inches vs. 15–18 inches for other strategies) and fertilizer (216–307 lbs vs. 450 lbs for farmer best practices). For operations prioritizing environmental impact or input cost reduction, MPC may be the preferred choice despite lower revenue.

7.2. Computational considerations

Several design choices were driven by computational tractability. First, for the MPC implementation, we use a two-tier temporal scheme: plant dynamics are simulated at hourly resolution (matching the crop model's physiological timescales), but decisions are made daily. This daily decision scale is both practically realistic and computationally necessary—solving 121 daily optimizations is tractable, while 2,900 hourly optimizations would be prohibitive. Second, the planning horizon is constrained to 3–14 days; longer horizons would increase the CFTOC problem size and solver time.

7.3. Limitations and Future Extensions

Several model limitations suggest directions for future work:

Growth model. The logistic equation (1) assumes symmetric growth around the inflection point. Richards growth generalizes this with a shape parameter v:

$$\frac{dx}{dt} = a_x x \left[1 - \left(\frac{x}{k_x}\right)^v\right] \tag{62}$$

where $v > 1$ produces steeper early growth and $n < 1$ produces steeper late growth.

Absorption kernels. Gaussian kernels are symmetric, but physiological absorption often exhibits fast activation followed by slow decay. Log-normal or Gamma kernels could better capture this asymmetry.

Saturation. The current model does not explicitly limit nutrient uptake. Modeling nutrient saturation would provide more realistic response to over-application.

Perfect forecast assumption. Our MPC formulation assumes perfect weather forecasts over the planning horizon. Incorporating forecast uncertainty through stochastic MPC or robust optimization would improve real-world applicability.

Single-point model. The crop model represents a single plant without spatial heterogeneity. Field-scale implementation would require accounting for spatial variation in soil properties, drainage, and microclimate.

Soil moisture dynamics. The current model treats irrigation as directly available water without explicitly modeling soil moisture content. A more realistic extension would incorporate soil water balance dynamics.

8. Conclusion

This paper presented a coupled ODE model for crop growth that captures two physiological phenomena often neglected in optimization studies: delayed nutrient absorption and cumulative stress tracking. The model tracks five state variables (plant height, leaf area, number of leaves, flower size, and fruit biomass) with time-varying growth rates modulated by nutrient factors derived from a series of transformations which capture the physiology of the stress and delayed absorption.

We compared two optimization approaches for irrigation and fertilizer scheduling: genetic algorithm (GA) optimization for fixed seasonal strategies and model predictive control (MPC) for adaptive daily decision-making. Applied to corn production in Iowa across 21 stochastic weather scenarios, both optimization-based methods outperform farmer best practices. GA optimization achieves 35% higher mean revenue than conventional practices ($842 vs. $626 per acre), demonstrating the value of computational optimization in agriculture.

Contrary to our initial hypothesis that adaptive control would outperform fixed strategies under weather uncertainty, the results show that MPC does

not outperform the GA strategies. GA (Normal) outperforms MPC in all 21 scenarios, while GA (Drought) outperforms MPC in 9 of 21 scenarios. The explanation lies in the mismatch between MPC's planning horizon (9 days) and the crop model's physiological timescales (fertilizer absorption spreads over 300+ hours). Although the 9 day horizon was found by Bayesian optimization to be optimal, MPC's daily adaptation cannot substitute for the GA's full-season optimization horizon when system dynamics (especially fertilizer absorption dynamics) are slow.

The practical conclusion is straightforward: for seasonal agricultural planning, pre-optimization for expected conditions outperforms adaptive control. If normal weather is expected, use GA (Normal); if drought is anticipated, use GA (Drought); if a different weather condition is expected, a GA can be tuned to that expectation. However, MPC offers a sustainability advantage worth considering: it achieves competitive returns while using dramatically less water and fertilizer than all other strategies, which may be valuable for operations prioritizing environmental impact or input cost reduction.

Future work should explore inclusion of soil nutrient effects, crop interaction with a field-scale model, and field validation of the model. Though this study calibrated model parameters to corn, the framework generalizes readily to other crops by adjusting the growth parameters and input-effect relationships.

9. Acknowledgements

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Declarations

Competing Interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Code availability The source code used for this study is archived on Zenodo at `https://doi.org/10.5281/zenodo.18204023`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process During the preparation of this work the authors used ChatGPT and Claude Code in order to generate some portions of the code base, though no underlying theory, and refine the original drafts of the paper. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Appendix A. Supplementary Information

This appendix provides additional figures that support the main results but are not essential to the primary narrative.

Appendix A.1. Nutrient absorption dynamics

The following figures illustrate the detailed nutrient absorption dynamics under the baseline farmer strategy (Section 6.2).

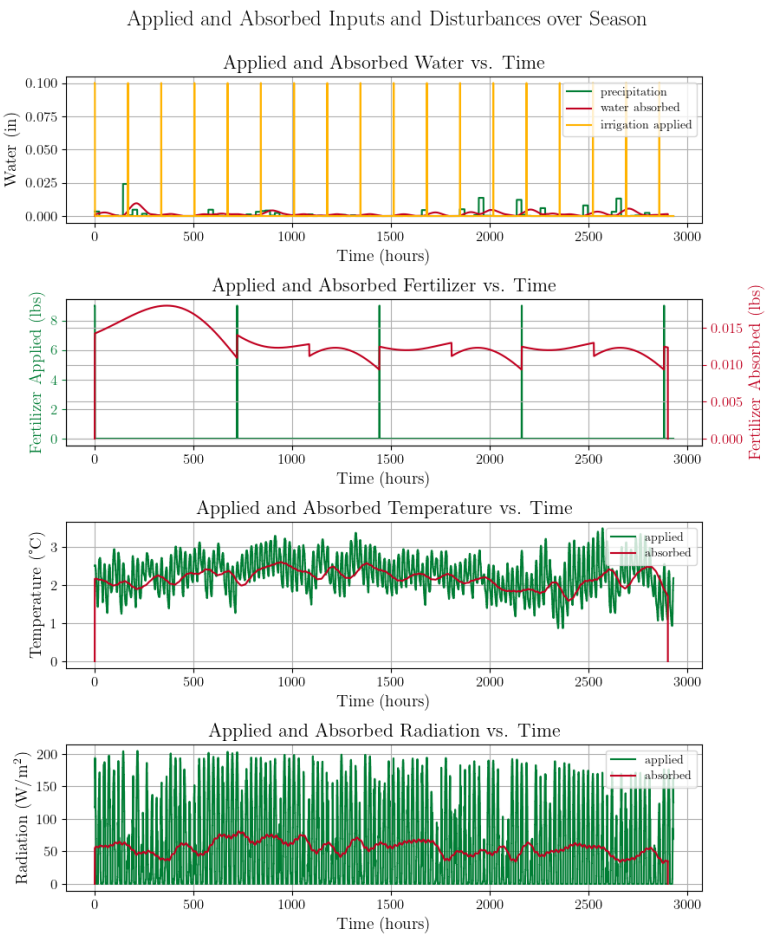

Figure A.11: Applied versus absorbed nutrients under the baseline farmer strategy. The delayed absorption dynamics are evident in the lag between applied inputs and the smoothed absorbed signals. Water absorption ($\sigma_w = 30$ hr) responds more quickly than fertilizer absorption ($\sigma_f = 300$ hr).

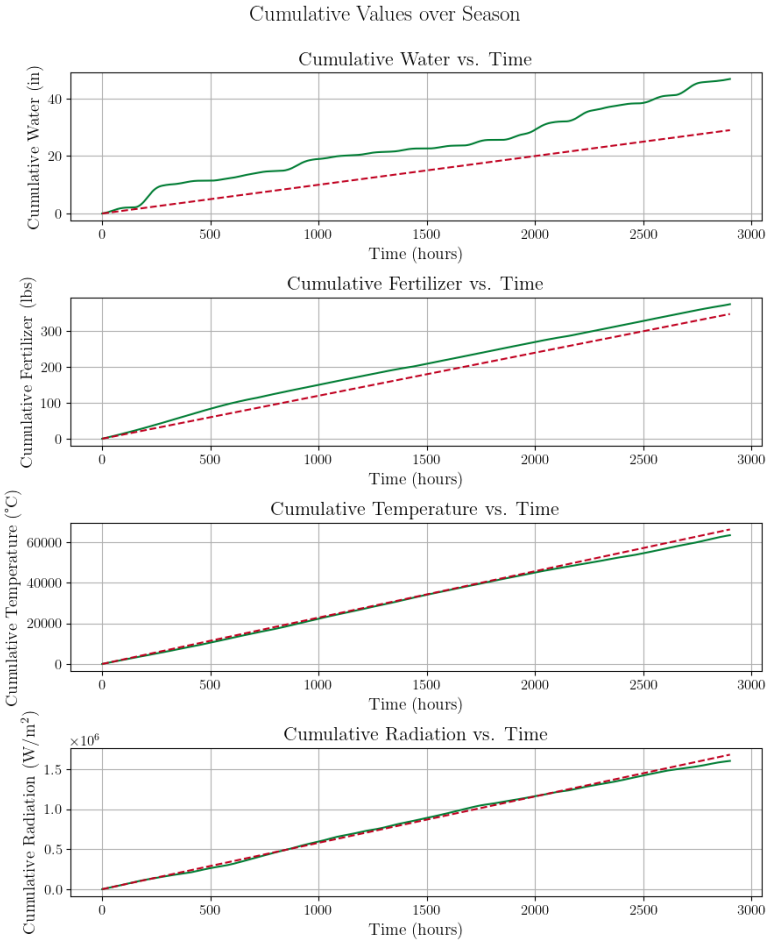

Figure A.12: Cumulative absorbed nutrients (solid) versus expected levels (dashed red). The growing gap between actual and expected water absorption reflects the drought stress accumulating over the season.

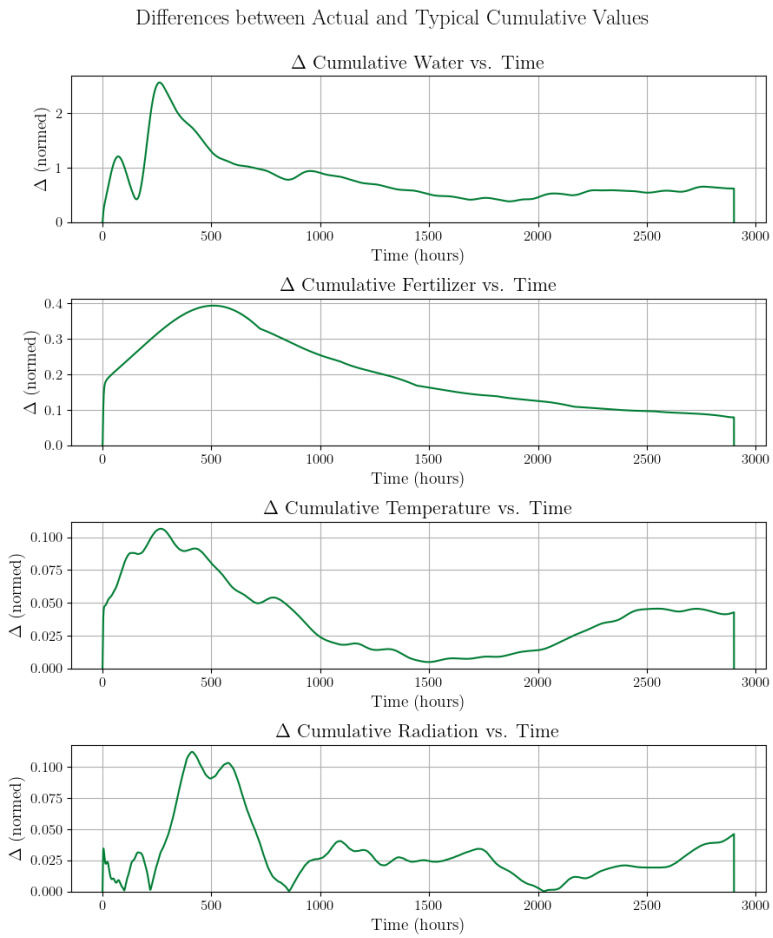

Figure A.13: Instantaneous divergence from expected cumulative nutrient levels. Higher divergence indicates greater stress.

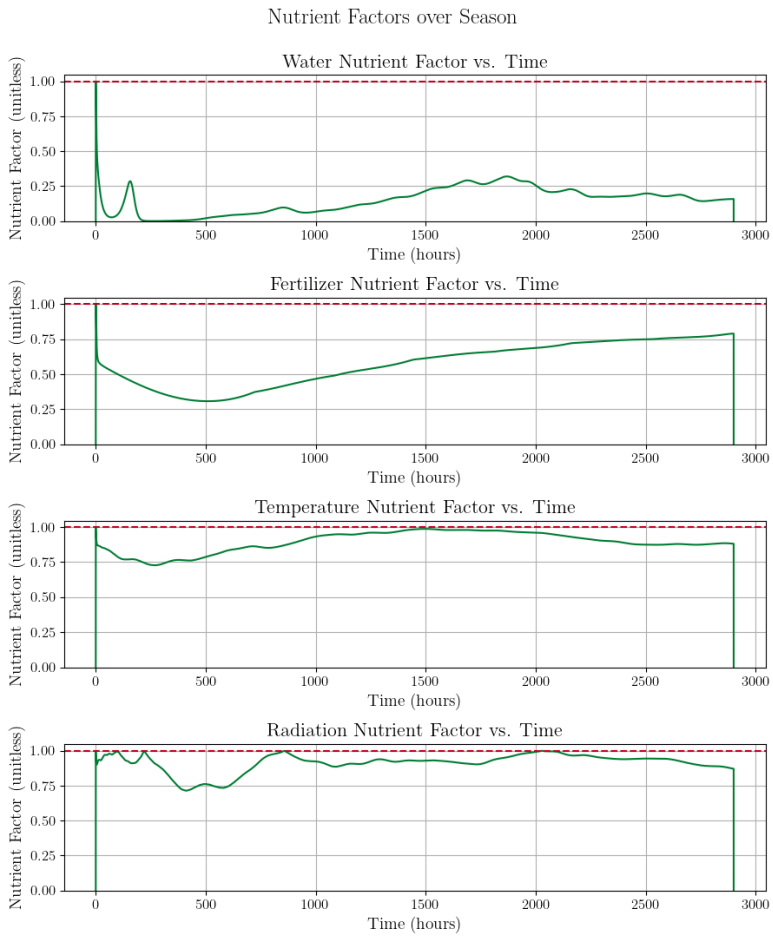

Figure A.14: Nutrient factor evolution under the baseline scenario. The water nutrient factor ν_w declines due to cumulative drought stress, while fertilizer, temperature, and radiation factors remain closer to 1.0 (no stress). The declining ν_w (which later recovers) reduces effective growth rates and carrying capacities throughout the season.

Appendix A.2. Suboptimal GA strategies

To verify that the model appropriately discriminates between effective and ineffective strategies, we examined the performance of suboptimal individuals in the GA population.

References

[1] Economic Research Service, Farm income and wealth statistics, US Department of Agriculture (2024).
URL https://www.ers.usda.gov/data-products/farm-income-and-wealth-statistics/

[2] T. I. Zohdi, A machine-learning enabled digital-twin framework for next generation precision agriculture and forestry, Computer Methods in Applied Mechanics and Engineering 428 (2024) 117250. doi:10.1016/j.cma.2024.117250.

[3] E. Mengi, C. J. Becker, M. Sedky, S. Yu, T. I. Zohdi, A digital-twin and rapid optimization framework for optical design of indoor farming systems, Computational Mechanics 72 (2023) 953–970. doi:10.1007/s00466-023-02421-9.

[4] P. Goodrich, O. Betancourt, A. Arias, T. I. Zohdi, Placement and drone flight path mapping of agricultural soil sensors using machine learning, Computers and Electronics in Agriculture 198 (2022) 107591. doi:10.1016/j.compag.2022.107591.

[5] I. Tagkopoulos, S. F. Brown, X. Liu, Q. Zhao, T. I. Zohdi, J. M. Earles, N. Nitin, D. E. Runcie, D. G. Lemay, A. D. Smith, P. C. Ronald, H. Feng, G. D. Youtsey, Special report: AI institute for next generation food systems (AIFS), Computers and Electronics in Agriculture 196 (2022) 106819. doi:10.1016/j.compag.2022.106819.

[6] P.-F. Verhulst, Notice sur la loi que la population suit dans son accroissement, Correspondance mathématique et physique 10 (1838) 113–126. doi:10.1007/BF02309004.

[7] J. W. Jones, G. Hoogenboom, C. H. Porter, K. J. Boote, W. D. Batchelor, L. Hunt, P. W. Wilkens, U. Singh, A. J. Gijsman, J. T. Ritchie, The

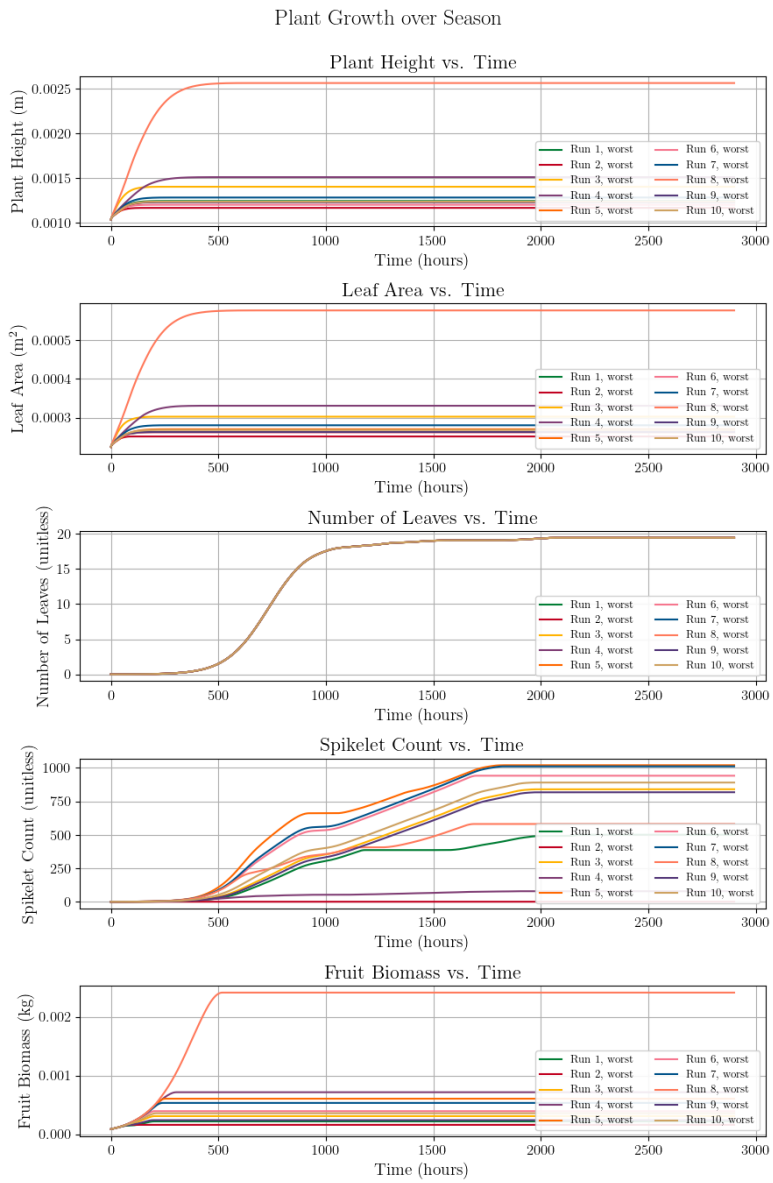

Figure A.15: Plant state variable trajectories for the worst member from each GA run's final population. These suboptimal strategies achieve almost no growth, confirming that the model does not guarantee excellent growth and appropriately penalizes poor resource allocation strategies.

dssat cropping system model, European Journal of Agronomy 18 (3-4) (2003) 235–265. doi:10.1016/S1161-0301(02)00107-7.

[8] D. P. Holzworth, N. I. Huth, P. G. deVoil, E. J. Zurcher, N. I. Herrmann, G. McLean, K. Chenu, E. J. van Oosterom, V. Snow, C. Murphy, et al., Apsim–evolution towards a new generation of agricultural systems simulation, Environmental Modelling & Software 62 (2014) 327–350. doi:10.1016/j.envsoft.2014.07.009.

[9] C. Van Diepen, J. Wolf, H. Van Keulen, C. Rappoldt, Wofost: a simulation model of crop production, Soil Use and Management 5 (1) (1989) 16–24. doi:10.1111/j.1475-2743.1989.tb00755.x.

[10] R. Gebbers, V. I. Adamchuk, Precision agriculture and food security, Science 327 (5967) (2010) 828–831. doi:10.1126/science.1183899.

[11] A. Singh, An overview of the optimization modelling applications, Journal of Hydrology 466–467 (2012) 167–182. doi:10.1016/j.jhydrol.2012.08.004.

[12] J. E. Epperson, J. E. Hook, Y. R. Mustafa, Dynamic programming for improving irrigation scheduling strategies of maize, Agricultural Systems 42 (1) (1993) 85–101, applications of Dynamic Optimization Techniques to Agricultural Problems. doi:https://doi.org/10.1016/0308-521X(93)90070-I. URL https://www.sciencedirect.com/science/article/pii/0308521X9390070I

[13] W. Ding, C. Lin, Application of genetic algorithms to agriculture: a review, Computers and Electronics in Agriculture 175 (2020) 105524. doi:10.1016/j.compag.2020.105524.

[14] D. E. Goldberg, Genetic Algorithms in Search, Optimization, and Machine Learning, Addison-Wesley, Reading, MA, 1989.

[15] R. Wardlaw, K. Bhaktikul, Application of genetic algorithms for irrigation water scheduling, Irrigation and Drainage 53 (4) (2004) 397–414. doi:10.1002/ird.121.

[16] Z. Yu, W. Fu, Optimization of nitrogen fertilization strategies for drip irrigation of cotton in large fields by dssat combined with a genetic algorithm, Applied Sciences 15 (7) (2025) 3580.

[17] IPCC, Climate change 2021: The physical science basis. contribution of working group i to the sixth assessment report of the intergovernmental panel on climate change, Tech. rep., Cambridge University Press (2021). doi:10.1017/9781009157896.

[18] J. B. Rawlings, D. Q. Mayne, M. Diehl, Model Predictive Control: Theory, Computation, and Design, 2nd Edition, Nob Hill Publishing, 2017.

[19] J. E. Sawyer, E. D. Nafziger, G. W. Randall, L. G. Bundy, G. W. Rehm, B. C. Joern, Nitrogen management for corn production, Iowa State University Extension Publication PM 1714 (2006).

[20] J. O. Payero, D. D. Tarkalson, S. Irmak, D. R. Davison, J. L. Petersen, Effect of irrigation amounts applied with subsurface drip irrigation on corn evapotranspiration, yield, water use efficiency, and dry matter production in a semiarid climate, Agricultural Water Management 95 (8) (2008) 895–908. doi:10.1016/j.agwat.2008.02.015.

[21] B. Sánchez, A. Rasmussen, J. R. Porter, Temperatures and the growth and development of maize and rice: a review, Global Change Biology 20 (2) (2014) 408–417. doi:10.1111/gcb.12389.

[22] A. Azi, et al., Growth and yield of sweet corn (zea mays var. rugosa) in response to bio-n and different levels of inorganic fertilizers, Asian Journal of Agricultural and Horticultural Research (2025).
URL https://files.sdiarticle5.com/wp-content/uploads/2025/02/Rev_AJAHR_130390_Azi_A.pdf

[23] M. S. Lum, N. F. b. Aldam, C. F. S. Chin, The effect of azolla (azolla pinnata) extract on the growth and yield of sweet corn under reduced nitrogen fertilization practice, Sarhad Journal of Agriculture 40 (Special Issue 1) (2024).
URL https://researcherslinks.com/current-issues/The-Effect-Azolla-Azolla-pinnata-Extract-Growth/14/34/10080/html

[24] Y. Liu, N. Wang, C. Jiang, Y. Wang, Effects of irrigation type and fertilizer application rate on growth, yield, and water and fertilizer use efficiency of silage corn in the north china plain, PeerJ 12 (2024) e18315. doi:10.7717/peerj.18315.
URL https://pmc.ncbi.nlm.nih.gov/articles/PMC11608015/

[25] M. A. Khan, K. B. Marwat, B. Gul, F. Wahid, H. Khan, S. Hashim, Pistia stratiotes l. (araceae): Phytochemistry, use in medicines, phytoremediation, biogas and management options, Pakistan Journal of Botany 46 (3) (2014) 851–860.
URL https://www.pakbs.org/pjbot/PDFs/46(3)/33.pdf

[26] J. Dodd, P. K. Sweby, S. Mayes, E. H. Murchie, A. S. Karunaratne, F. Massawe, M. J. Tindall, A multiscale mathematical model describing the growth and development of bambara groundnut, Journal of Theoretical Biology 560 (2023) 111373. doi:https://doi.org/10.1016/j.jtbi.2022.111373.

[27] R. C. Lewontin, D. Cohen, On population growth in a randomly varying environment, Proceedings of the National Academy of Sciences 62 (4) (1969) 1056–1060. doi:10.1073/pnas.62.4.1056.

[28] H. Ren, Y. Liu, J. Zhang, D. Li, Y. Wang, J. Zhang, Effects of agronomic traits and climatic factors on yield and yield stability of summer maize in the huang?huai?hai plain, Frontiers in Plant Science 13 (2022) 1042107. doi:10.3389/fpls.2022.1042107.
URL https://pmc.ncbi.nlm.nih.gov/articles/PMC9707337/

[29] C. J. Stubbs, D. D. Cook, D. J. Robertson, The effect of plant height on maize stalk lodging resistance, Field Crops Research 300 (2023) 109008. doi:10.1016/j.fcr.2023.109008.
URL https://par.nsf.gov/servlets/purl/10545305

[30] Y. Wang, J. Bao, X. Wan, S. Wu, C. Feng, Y. Qin, Y. Guo, Z. Dong, Genetic structure and molecular mechanisms underlying the formation of tassel, anther, and pollen in the male inflorescence of maize (zea mays l.), Cells 11 (11) (2022) 1753. doi:10.3390/cells11111753.
URL https://pmc.ncbi.nlm.nih.gov/articles/PMC9179574/

[31] D. L. Silveira, A. Cargnelutti Filho, M. Toebe, C. Burin, V. J. Szareski, Adaptability and stability of grain yield and maize tassel traits, Ciencia Rural 54 (3) (2024) e20230319. doi:10.1590/0103-8478cr20230319. URL https://www.scielo.br/j/cr/a/5tGN99hwhFgDjGmx8WgfDns/?format=html&lang=en

[32] US Department of Agriculture, 2023 acreage data as of august 9, 2023, Fsa crop acreage data reported to fsa, US Department of Agriculture Farm Service Agency (2023).

[33] M. Sengupta, Y. Xie, A. Lopez, A. Habte, G. Maclaurin, J. Shelby, The national solar radiation data base (nsrdb), Renewable and Sustainable Energy Reviews 89 (2018) 51–60. doi:10.1016/j.rser.2018.03.003.

[34] National Oceanic and Atmospheric Administration, Climate data online (2024).
URL https://www.ncdc.noaa.gov/cdo-web/

[35] J. Sawyer, E. Nafziger, G. Randall, L. Bundy, G. Rehm, B. Joern, Concepts and rational for regional nitrogen rate guidelines for corn, University Extension Iowa State University Ames (2006).

[36] L. J. Abendroth, R. W. Elmore, M. J. Boyer, S. K. Marlay, In corn growth and development, Iowa State University Extension Publication PMR 1009 (2011).

[37] F. Schierbaum, Book reviews: Corn: Chemistry and technology, 2nd edition. by pamela j. white and lawrence a. johnson (editors), Starch-starke - STARCH 56 (2004) 263–264. doi:10.1002/star.200490027.

[38] Bayer Crop Science, Height difference between traditional corn products and short stature corn hybrids (2018).
URL https://www.cropscience.bayer.us/articles/traits/height-difference-between-traditional-corn-products-and-short-stature-corn-hy

[39] Keller's Farmstand, Sweet corn vs. field corn height (2018).
URL https://kellersfarmstand.com/sweet-corn-vs-field-corn-height/

[40] D. Nanda, Number of corn leaves provides insight into maturity (2017).
URL https://www.farmprogress.com/corn/number-of-corn-leaves-provides-insight-into-maturity

[41] R. L. Nielsen, Tassel emergence and pollen shed (2025).
URL https://www.agry.purdue.edu/ext/corn/news/timeless/tassels.html

[42] Urban Farmer Seeds (UF Seeds), Pencil cob corn seed (2024).
URL https://www.ufseeds.com/product/pencil-cob-corn-seed/COPCB.html

[43] Bayer Crop Science, Corn growth stages and gdu requirements (2020).
URL https://www.cropscience.bayer.us/articles/bayer/corn-growth-stages-and-gdu-requirements

[44] University of California, Davis, Crop development (2024).
URL https://corn.ucdavis.edu/crop-development

[45] A. J. Sisson, D. Mueller, S. P. Conley, C. K. Gerber, Corn growth stages (2020).
URL https://cropprotectionnetwork.org/web-books/crop-scouting-basics-for-corn-and-soybean?section=21-corn-growth-stages

[46] Western Cooperative Company, Corn growth (2024).
URL https://www.westco.coop/pages/custom.php?id=33409

[47] Corn States, Corn growth stages and gdu requirements (2018).
URL https://www.corn-states.com/app/uploads/2018/07/corn-growth-stages-and-gdu-requirements.pdf

[48] J. Eisenthal, Fertilizer prices put economic squeeze on farmers (2025).
URL https://www.farms.com/news/fertilizer-prices-put-economic-squeeze-on-farm
aspx

[49] National Corn Growers Association, What is driving high production costs in corn? (2025).
URL https://www.morningagclips.com/what-is-driving-high-production-costs-in-corn/

[50] K. Swanson, High production cost series, part 3 (2025).
URL https://ncga.com/stay-informed/media/the-corn-economy/article/2025/08/high-production-cost-series-part-3

[51] Farmonaut, Average cost to clear land, plant corn, farm insurance (2025).
URL https://farmonaut.com/usa/average-cost-to-clear-land-plant-corn-farm-insu

[52] M. Newman, Irrigating more u.s. crops by mid-century will be worth the investment (2023).
URL https://natsci.msu.edu/news/2023-08-irrigating-more-us-crops-by-mid-centu aspx

[53] L. S. Rathore, M. Kumar, R. T. McNider, N. Magliocca, W. Ellenburg, Contrasting corn acreage trends in the midwest and southeast: The role of yield, climate, economics, and irrigation, Journal of Agriculture and Food Research 18 (2024) 101373. doi:https://doi.org/10.1016/j.jafr.2024.101373.
URL https://www.sciencedirect.com/science/article/pii/S2666154324004101

[54] A. Berger, Estimating bushels of corn on the ground by counting ears prior to grazing with cattle (2020).
URL https://beef.unl.edu/estimating-bushels-corn-ground-counting-ears-prior-g

[55] B. Groenke, Estimating corn yield potential (2018).
URL https://www.canr.msu.edu/news/estimating-corn-yield-potential

[56] Bayer Crop Science, Estimating corn yield (2023).
URL https://www.cropscience.bayer.us/articles/channel/estimating-corn-yield

[57] M. Licht, Z. Clemens, Corn yield estimates (2020).
URL https://crops.extension.iastate.edu/encyclopedia/corn-yield-estimates

[58] USDA Agricultural Marketing Service, Iowa daily cash grain bids (2026).
URL https://www.ams.usda.gov/mnreports/ams_2850.pdf

[59] E. Sucu, Optimizing plant density improves corn silage quality (2020).
URL https://dellait.com/optimizing-plant-density-improves-corn-silage-quality

[60] D. S. Dhaliwal, Y. Assefa, M. M. Williams, Historical trends in sweet corn plant density tolerance, Frontiers in Plant Science 12 (2021) 685066. doi:10.3389/fpls.2021.685066.
URL https://pmc.ncbi.nlm.nih.gov/articles/PMC8492913/

[61] Hanna Instruments, Weight of corn (2024).
URL https://www.hanna-one.com/weight-corn.html

[62] M. I. Hommood, M. Isaak, Physical and mechanical properties of sweet corn plant, Agricultural Engineering International: CIGR Journal 22 (2) (2020) 138–147.
URL https://cigrjournal.org/index.php/Ejounral/article/download/5306/3445/0

[63] W. L. Kranz, S. Irmak, S. J. van Donk, C. D. Yonts, D. L. Martin, Irrigation management for corn, Tech. Rep. G1850, University of Nebraska–Lincoln Extension (2008).
URL https://extensionpubs.unl.edu/publication/g1850/

[64] J. L. Hatfield, J. H. Prueger, Temperature extremes: Effect on plant growth and development, Weather and Climate Extremes 10 (2015) 4–10. doi:10.1016/j.wace.2015.08.001.

[65] M. L. Bynum, G. A. Hackebeil, W. E. Hart, C. D. Laird, B. L. Nicholson, J. D. Siirola, J.-P. Watson, D. L. Woodruff, Pyomo—Optimization Modeling in Python, 3rd Edition, Springer, 2021. doi:10.1007/978-3-030-68928-5.

[66] A. Wächter, L. T. Biegler, On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming, Mathematical Programming 106 (1) (2006) 25–57. doi:10.1007/s10107-004-0559-y.

[67] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, N. de Freitas, Taking the human out of the loop: A review of Bayesian optimization, Proceedings of the IEEE 104 (1) (2016) 148–175. doi:10.1109/JPROC.2015.2494218.

[68] J. Bergstra, R. Bardenet, Y. Bengio, B. Kégl, Algorithms for hyper-parameter optimization, in: J. Shawe-Taylor, R. Zemel, P. Bartlett, F. Pereira, K. Weinberger (Eds.), Advances in Neural Information Processing Systems, Vol. 24, Curran Associates, Inc., 2011.
URL https://proceedings.neurips.cc/paper_files/paper/2011/file/86e8f7ab32cfd12577bc2619bc635690-Paper.pdf

[69] B. Davies, J. A. Coulter, P. H. Pagliari, Timing and rate of nitrogen fertilization influence maize yield and nitrogen use efficiency, PLOS ONE 15 (5) (2020) e0233674. doi:10.1371/journal.pone.0233674.

[70] B. Gouthami, B. Ashok Kumar, Water management in maize, Just Agriculture 2 (7), article ID: 059, e-ISSN: 2582-8223 (March 2022).
URL https://justagriculture.in/files/newsletter/2022/march/059.pdf

[71] R. Roy, A. Finck, G. Blair, H. Tandon, Nutrient management guidelines for some major field crops, in: Plant Nutrition for Food Security: A Guide for Integrated Nutrient Management, no. 16 in FAO Fertilizer and Plant Nutrition Bulletin, Food and Agriculture Organization of the United Nations, Rome, 2006, Ch. 8, pp. 235–274, fAO document a0443e.
URL https://www.fao.org/4/a0443e/a0443e04.pdf

[72] A. Mallarino, P. Barbieri, K. Pecinovsky, Long-term phosphorus and potassium fertilization effects on yields of corn and soybean grown in rotation, Tech. Rep. ISRF14-13, RFR-A14104, Iowa State University, Northeast Research and Demonstration Farm, Nashua, IA, winter 2015 (2015).
URL https://www.agronext.iastate.edu/soilfertility/info/NE%20farm%20report_PK%20Winter%202015-Long%20term%20P-K%20effects.pdf

[73] A. Robinson, Usda crop report continues large 2025 crop production estimates, https://www.iowafarmbureau.com/Article/USDA-Crop-Report-Continues-Large-2025-Crop-Production-Estimates, iowa Farm Bureau. Accessed: 2026-03-08 (2025).

[74] USDA Agricultural Marketing Service, Iowa daily cash grain bids, https://www.ams.usda.gov/mnreports/ams_2850.pdf, live-

stock, Poultry and Grain Market News Division. Accessed: 2026-03-08 (2026).

[75] K. Swanson, High production cost series, part 3, https://ncga.com/stay-informed/media/the-corn-economy/article/2025/08/high-production-cost-series-part-3, national Corn Growers Association. Accessed: 2026-03-08 (2025).